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# Urban shading and the solar potential of vehicle-integrated photovoltaics

## On-board irradiance measurements in Lisbon

Internship report, April–August 2026

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# Abstract

Vehicle-integrated photovoltaics can supply part of an electric vehicle's energy where that energy is consumed, but in a city the roof of a car spends much of its time in the shadow of buildings. This work, carried out at the Instituto Dom Luiz (Faculdade de Ciências, Universidade de Lisboa), set out to quantify that shading loss from measurements collected by irradiance sensors carried by vehicles in Lisbon.

The loss of part of the campaign data made the intended estimate impossible. What the internship produced instead is the chain that will allow it to be made: a documented path from the raw sensor export to the interpreted figure, covering decoding, normalisation, spatial joining, cleaning and the construction of specialised subsets, held together by a file-naming convention that lets any figure be traced back to the data behind it. A second measurement campaign was launched and brought into that chain.

Three results hold on the surviving data. The movement status of the vehicle is the robust axis: relative losses are lower while driving than while parked in 93 % of the comparable bins, under every filter tested. Mean building height within 50 m is a usable predictor of loss, with a factor of two between the most open and the most built-up surroundings. The satellite irradiance reference was validated against a ground pyranometer before being used, and its disagreement with the on-board sensor was shown to be the shading signal itself rather than a defect. Conversely, the urban/rural contrast that organised earlier work is not testable on this dataset, and the class-level analysis suggests it stood as a proxy for the openness of the sky rather than as a cause.

**Keywords:** vehicle-integrated photovoltaics; urban shading; solar irradiance; data processing chain; land cover.

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# Abbreviations and symbols

**BH** building height (file-name segment `_bh`)

**BHI** beam horizontal irradiance (W/m$^2$)

**BSRN** Baseline Surface Radiation Network

**CAMS** Copernicus Atmosphere Monitoring Service

**CLC** CORINE Land Cover

**COS** Carta de Uso e Ocupação do Solo (Portuguese land-cover map)

**CSV** comma-separated values (file format)

**CVE** average electric-vehicle consumption (Portuguese *consumo médio de um veículo elétrico*)

**DGT** Direção-Geral do Território (Portuguese land survey)

**DHI** diffuse horizontal irradiance (W/m$^2$)

**DNI** direct normal irradiance (W/m$^2$)

**EDR** extended daily range (km/(day kWp))

**FCUL** Faculdade de Ciências, Universidade de Lisboa

**GADM** Database of Global Administrative Areas

**GBA** GlobalBuildingAtlas

**GHI** global horizontal irradiance (W/m$^2$)

**GPS** global positioning system

**IDL** Instituto Dom Luiz

**IEA PVPS** International Energy Agency Photovoltaic Power Systems Programme

**LiDAR** light detection and ranging

**LoD1** level of detail 1 (prismatic building model)

**MMU** minimum mapping unit

**OSM** OpenStreetMap

**QC** quality control

**RMSE** root-mean-square error

**SVF** sky view factor

**SW** street width (file-name segment `_sw`)

**TEDS** ecological transition and sustainable development (French: *transition écologique et développement soutenable*)

**VIPV** vehicle-integrated photovoltaics

**VKT** vehicle-kilometres travelled

$G$ global horizontal irradiance (W/m$^2$)

$G_m$ irradiance measured by the on-board sensor (W/m$^2$)

$k_t$ clearness index, GHI/GHI$_{\text{extra}}$

$k_t'$ modified clearness index, corrected for air mass

$k_d$ diffuse fraction, DHI/GHI

$k_c$ clear-sky index, GHI/GHI$_{\text{clear}}$

$H$ mean height of the buildings within 50 m of the measurement (m)

$W$ width of the street at the measurement (m)

$H/W$ urban canyon aspect ratio

$\alpha$ obstruction angle subtended by the surrounding buildings (°)

$\gamma_s$ solar elevation angle, column `sun_elev` (°)

$\psi_s$ solar azimuth angle, column `sun_azim` (°)

# Glossary

**Urban canyon** The idealisation of a street as an infinitely long corridor of width $W$ between two continuous building rows of height $H$. The single ratio $H/W$ then fixes how much of the sky dome a point in the street can see, and with it the share of the diffuse radiation it receives. Used here to compare a measured loss against a theoretical one (Section 2.5.4).

**Pyranometer** An instrument that measures the global irradiance arriving on a horizontal surface, over the whole sky dome and the whole solar spectrum. The reference used in this work is the EKO MS-802 on the FCUL rooftop, a fixed station with an open sky, against which the satellite product is validated (Section 3.3).

**Shading factor** The fraction of the reference irradiance that never reaches the vehicle, that is, what the built environment intercepts. It is defined against a reference that is blind to local obstruction, which is what makes the difference between the two a measure of shading rather than of weather.

**Sky view factor** The fraction of the sky dome visible from a point, between 0 and 1. Under the isotropic assumption used here, the diffuse radiation received is proportional to it.

# 1 Introduction

## 1.1 VIPV in the decarbonisation of mobility

The electrification of the European vehicle fleet is advancing faster than the grid that has to supply it: if every driver plugs in on returning home, the evening peak would require major reinforcement of the distribution network. Photovoltaic modules integrated into the body of the vehicle, vehicle-integrated photovoltaics (VIPV), are one answer: they produce electricity where it is consumed, and SolarMoves puts the saving at up to 27 TW h per year by 2030, with a deferred grid investment of 20 % to 25 % [1].

They also extend the range available between two charging events. This extended daily range (km/(day kWp)) (EDR) reaches 17 km/(day kWp) to 18 km/(day kWp) on the roads of Lisbon, about one third of the annual charging demand of a car driven 30 km a day [2]; Pereira measured 9.1 km/(day kWp) for the municipality itself [3].

## 1.2 Research context

This work was carried out at the Instituto Dom Luiz (IDL), Faculdade de Ciências, Universidade de Lisboa, under the supervision of Miguel Centeno Brito. It continues two master's dissertations from the same group on the *Carros Solares* citizen-science campaign: Ferreira [4] set it up, Pereira [3] characterised the sensors and analysed the resulting potential. The present work reuses their sensors and reworks their figures.

The same group provides the theoretical counterpart: Brito et al. mapped the shading of the urban fabric of Lisbon from a LiDAR digital surface model [2], then generalised the result to a hundred cities through an urban-canyon model [5]. That is the model this work applies to its own measurements. The international frame is IEA PVPS Task 17, which brings together the SolarMoves consortium [1] and the urban-canyon work at UPM Madrid.

## 1.3 Urban shading

Two very different phenomena reduce the irradiance a vehicle receives. Clouds attenuate the resource itself, over kilometres and minutes; buildings, trees and street furniture intercept the beam and hide part of the sky dome, over metres, with a geometry that changes as the vehicle moves. The reference against which losses are measured, a satellite product or a ground pyranometer, reproduces the first and is blind to the second. The shading factor, the fraction of the reference irradiance that never reaches the vehicle, is therefore built on what the reference cannot see.

Brito et al. report median annual losses of 24 % to 28 % on the roads of Lisbon and 52 % on its

parking spaces [2], which their canyon model translates into a range reduction of 18 % for driving vehicles against 60 % for parked ones [5]; SolarMoves measures an energy-weighted shading factor of 19.4 % in urban environments, rising to 63 % for a car left parked in the shade [1]. Two campaigns have placed the figure at 35 % [4] and then between 18 % and 25 % [3], a spread that is methodological rather than physical; and beyond about 40 %, over an eight-year lifetime against the German grid mix, on-board generation stops beating grid charging on emissions [6].

## 1.4 Campaigns

Both campaigns share the same instrument: the Solar Jinie, a small self-powered box magnetically fixed to the vehicle roof, recording one row per minute in daylight. Four units are involved, J30 to J33.

The first, referred to throughout as the *standard* campaign, had the sensors carried by the vehicles of volunteers. J30 and J31 began on 20 December 2021 with a test and rooftop calibration phase at the FCUL reference station, switched to vehicle operation on 15 June 2022, and end on 25 October 2022 and 21 May 2023 respectively. Everything recorded from June 2023 onwards has since been lost. The two surviving files stop at those dates, J30 over a shorter period than J31. The point is therefore no longer an absolute figure but the chain that will allow one to be produced later, on a sample whose representativeness is not established.

The second campaign was launched during this internship, with the same sensors in an inverted setting: tourist tuk-tuks driving through the day in the dense historic centre, along repeated itineraries. This is the vehicle category Brito et al. identify as the most relevant short-term market for VIPV [5]. It started on 29 June 2026 with a single vehicle, and 24 days of data are available. A poster is displayed on board for the passengers (Chapter H).

## 1.5 Data used

Each measurement has to be placed against a reference irradiance and described by the environment it was taken in. The reference irradiance comes from the radiation service of the Copernicus Atmosphere Monitoring Service (CAMS), part of Copernicus, the European Union's Earth-observation programme [7], and the solar position from `pvlib`, the open-source Python library maintained by the photovoltaic community.

The environment comes from four more. Carta de Uso e Ocupação do Solo (COS), the national land-cover map of Portugal published by the Direção-Geral do Território (DGT), resolves the urban fabric; CORINE Land Cover (CLC), its European counterpart from the Copernicus Land Monitoring Service, keeps the results comparable beyond it. Building heights come from GlobalBuildingAtlas (GBA), where Zhu et al. derive footprints and heights from satellite imagery by machine learning [8], reached through the U-Shift IMPT-data release centroid file. Street widths come from OpenStreetMap, whose network carries width as an explicit tag, a lane count, or the road class alone. Chapter A gives the origin, resolution and licence of each.

# 2 Methods

## 2.1 Pipeline architecture and conventions

The pipeline is a set of independent Python scripts rather than a single program. Each one reads one or more CSV files, appends columns or removes rows, and writes a new CSV. None modifies its input, so every intermediate state stays on disk and can be inspected.

The file name is what holds the chain together. No script names its output from scratch: each appends its own suffix to the name it received. A name therefore records the exact sequence of treatments applied to it, with no external register to consult, and any figure can be traced back to the subset that produced it. Chapter C gives the meaning of each segment and follows one file through the whole chain.

The two campaigns keep separate join chains, and therefore separate file stems, but the analysis code above them is shared. Every script resolves its input paths from three parameters set at the top of the file: the campaign, the sensor and the active suffix. Switching an analysis from one variant to another, or from one campaign to the other, is a one-line edit. That sharing is possible because the column schema is made identical upstream, at the normalisation stage of Section 2.2: measurement columns are sensor-prefixed in both campaigns (`J31 lat`, `J31 Rad calib`), while joined columns carry fixed names (`ghi`, `sun_elev`, `land_use_clcv`). Each script declares the columns it requires and fails immediately if one is absent.

## 2.2 Decoding and normalisation

The tuk-tuk campaign does not deliver the same raw material as the standard one. Its export is one row per transmitted packet, each carrying a JSON payload that holds five delta-encoded measurements, so decoding begins by expanding every packet into its individual minutes. Chapter C describes the encoding and the fields recovered.

Two consequences matter here. A per-sensor linear calibration $\mathrm{rad}_{\mathrm{calib}} = a\,\mathrm{rad}_{\mathrm{raw}} + b$ is applied at this stage, and until the post-campaign calibration session is held $a = 1$ and $b = 0$. Every tuk-tuk value in this report is therefore expressed in raw sensor units, and will shift when the true coefficients are measured. Normalisation then renames the decoder's generic column names into the sensor-prefixed schema of the standard campaign, which is what allows a single body of analysis code to serve both campaigns.

## 2.3 Cleaning process and dataset variants

Cleaning is a stack of filters, each optional except the first, each producing a file of its own, so that no analysis has to trade sample size against data quality: it picks the level it needs.

An optional *start-date* step discards everything before the day the sensors left the calibration rooftop, 15 June 2022 for the standard campaign and 29 June 2026 for the tuk-tuk one, and marks the downstream family with `_sd`. The mandatory step then removes invalid rows: any record with a missing required field, a zero raw or calibrated reading, a calibrated irradiance below the $-4\,\mathrm{W/m^2}$ physical floor of the BSRN recommended tests [9], or a land-cover code outside the valid CLC range.

The optional "high-quality" step keeps only the minutes whose radiometry and geometry are sound enough to carry a loss: solar elevation above 10°, global irradiance above $25\,\mathrm{W/m^2}$, and two quality-control flags of Perez-Astudillo et al. [10] that a single-component sensor can support, the upper physical bound and a modified clearness index $k'_t > 1$, physically impossible, where $k'_t$ is the clearness index corrected for air mass [11, 12].

The last step keeps only the minutes for which $\mathrm{GHI}/\mathrm{GHI}_{\mathrm{clear}} \geq 0.95$. Under a clear sky the cloud attenuation is negligible, so any shortfall of the measured irradiance below the reference can be attributed only to local built shading.

Two geographic filters can then be applied, and they do not chain: strict containment in the Lisbon municipality polygon of the GADM level-2 boundaries, or a bounding box over Lisbon and Cascais. The resulting extents are nested, `cleaned` ⊇ `region` ⊇ `lisbon`. A time-range selector, working on whole days or to the minute, can finally be applied to any of them (Figure 2.1).

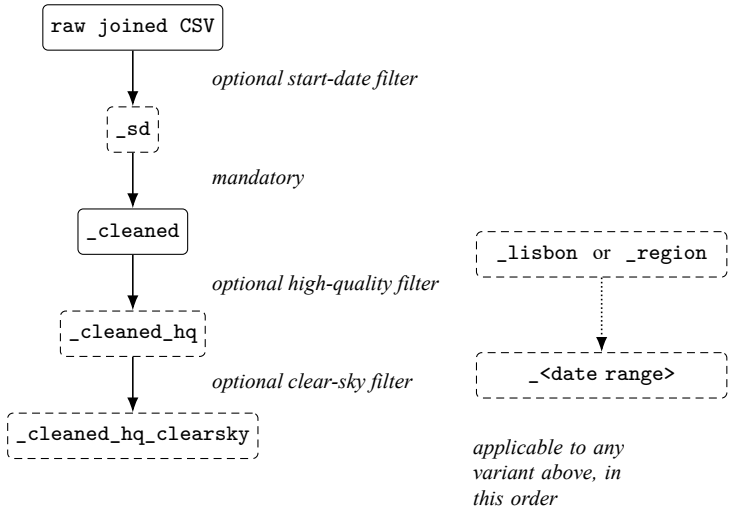


**Figure 2.1:** Dataset variants produced by the cleaning pipeline. Solid outlines mark the mandatory path, dashed ones the optional steps. The quality and clear-sky filters apply in sequence, the second on the output of the first. The two geographic filters are alternatives, not successive stages; the time-range selector may be applied to any variant.

## 2.4 Joins

Every external dataset is joined once, on the raw file, before any cleaning. Joining after cleaning would repeat the same spatial computation for every quality level, and would make the variants incomparable if a join were ever changed between two runs. Each join appends columns and never drops, adds or reorders a row, so the file grows in width and stays aligned with its input line for line. Chapter C gives the order of the stages, the columns each one contributes, and the implementation notes.

Reference irradiance comes from two sources. CAMS is joined by snapping each measurement to the nearest node of a 0.05° grid, downloading one time series per occupied node at one-minute resolution, and interpolating it onto the measurement timestamps; the nodes are pooled across sensors so that no series is downloaded twice. CAMS delivers accumulated irradiation in W h/m$^2$ per interval, converted to W/m$^2$ by dividing by the interval duration. The tuk-tuk campaign stays within Lisbon, close to the FCUL rooftop pyranometer, so that instrument is joined as well, by linear interpolation in time and without extrapolation beyond the range it covers. Solar elevation and azimuth are computed with `pvlib` at the true GPS coordinates of each measurement.

Land cover follows, by exact point-in-polygon test against the COS2023 and CLC vector layers, one polygon per point or an explicit unknown. The CLC vector join replaces an earlier raster-derived column, which is left in the files but no longer read. The replacement was checked against a second recomputation that samples the CLC raster at the same coordinates: the two recomputations agree on 98 to 99 % of points, while the old column agrees with neither as consistently. The comparison was also made on an interactive diagnostic map written for the purpose.

Building height is joined differently. What shades a street is the buildings on both of its sides, not the single nearest one, so the join is a buffer: every building centroid within 50 m of the point is collected, and their mean height, maximum height and count are appended (Figure 2.2). A count of zero is a legitimate result, an open space or a wide road, and is kept distinct from a missing value.

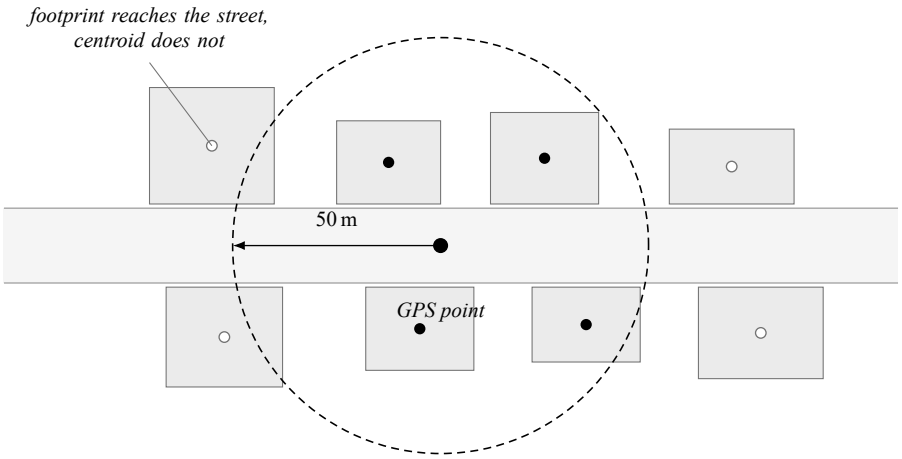


**Figure 2.2:** How building height is attached to a measurement. Every building centroid inside the 50 m buffer, filled here, is collected; those outside it, hollow, are ignored. In this example four of the eight buildings are counted, and $H$ is the mean of their four heights.

Street width comes last, from the OpenStreetMap network. The nearest street edge to each point is found, and its width estimated by a three-tier fallback: an explicit `width` tag, failing that the lane count multiplied by a standard lane width, failing that a default per road class. Which of the three was used is recorded alongside the value for every point, so any figure built on this column can state how much of it rests on a tagged width rather than on an assumption. The estimate is a street width, not the facade-to-facade distance the canyon model calls for. That model, set out in Section 2.5.4, idealises a street as a corridor of width $W$ between two continuous building rows of height $H$, and derives the shading from the single ratio $H/W$. Taking the street width for $W$ underestimates the true width, so the theoretical loss computed from it is a conservative upper bound.

## 2.5 Computational methods

### 2.5.1 Movement classification and distance measurement

The distance between two positions is a great-circle distance on a sphere, computed with the haversine formula. Writing $\varphi$ for latitude and $\lambda$ for longitude in radians,

$$d = 2R \arcsin \sqrt{\sin^2\left(\frac{\Delta\varphi}{2}\right) + \cos\varphi_1 \cos\varphi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right)}, \tag{2.1}$$

with $\Delta\varphi = \varphi_2 - \varphi_1$, $\Delta\lambda = \lambda_2 - \lambda_1$ and $R = 6{,}371$ km, the mean Earth radius. This distance serves two unrelated purposes in the pipeline.

The first is to decide, minute by minute, whether the vehicle was driving or standing. Each consecutive pair of positions is compared: if the distance covered exceeds a threshold, the first point is labelled *moving*, otherwise *parked*. Two guards make this reliable. Pairs straddling an overnight gap are discarded rather than measured, since a sensor switched off in one place and on again in another would register a displacement it never travelled; without that guard, 57.9 % of night transitions would be classified as moving. Pairs separated by more than two minutes are dropped for the same reason.

The distance threshold is 30 m. A sensitivity sweep places the start of the plateau near 20 m, but the classification is more stable at 30 m, which over the one-minute sampling interval corresponds to 1.8 km/h.

The second use is a plain distance to a fixed point, the FCUL reference station, which supports the ring analyses of the results chapter. Here only the two coordinates matter and no temporal guard applies.

### 2.5.2 Loss and autonomy metrics

Every minute of measurement is turned into an energy, and every energy into a driving range. Writing $G$ for the reference irradiance from CAMS, $G_m$ for the calibrated on-board measurement and $\Delta t$ for the sampling interval, the energy actually collected per unit of installed peak power and the irradiance lost to shading are

$$E_{\text{solar}} = G_m \, \Delta t \qquad \text{and} \qquad \text{Loss}_{\text{irr}} = \max(0,\ G - G_m)\, \Delta t. \tag{2.2}$$

The truncation at zero in $\text{Loss}_{\text{irr}}$ is a deliberate asymmetry. The sensor regularly reads above the satellite estimate, because of the curvature of the roof, of reflections, or of the spatial averaging inherent to a satellite pixel, and those minutes are counted as zero loss rather than as negative loss. The metric is a shading loss, not a signed difference, and it cannot be negative by construction.

The solar autonomy, the driving range that energy provides, follows from a single conversion constant, the average electric-vehicle consumption:

$$\text{Autonomy} = \frac{E_{\text{solar}}}{\text{CVE}}, \qquad \text{Loss}_{\text{auto}} = \frac{\text{Loss}_{\text{irr}}}{\text{CVE}}, \tag{2.3}$$

both in km/kWp, with $\text{CVE} = 16.6$ kW h per 100 km, the certified average of the battery electric vehicles registered in the European Union [13]. The same notation is used in the two earlier dissertations of this group [3, 4]. The same expression applied to $G$ instead of $G_m$ gives the range the same vehicle would have achieved with no shading at all, and the gap between the two is what the loss figures report. Only the window between 06:00 and 21:00 local time is retained for the hourly plots, and any bin holding fewer than 600 minutes is marked as unreliable on the figures.

### 2.5.3 Aggregation

Absolute losses are additive and pose no question. The relative loss does, because a percentage has to be formed at some point, and where that point falls changes the answer.

Forming the ratio on every individual minute and averaging the ratios is the intuitive choice, and it is misleading: a minute at dawn contributing almost no energy but showing a 90 % relative loss weighs as much as a minute at noon carrying a hundred times the energy. The scheme retained throughout this work, labelled *method A* on the figures, pools every minute of the bin first and divides once,

$$\text{Loss}_{\text{rel}} = \frac{\sum \text{Loss}_{\text{auto}}}{\sum \text{Loss}_{\text{auto}} + \sum \text{Autonomy}} \times 100, \tag{2.4}$$

so that each minute is weighted by its own irradiance and the result is the fraction of the available energy that was lost, which is the quantity a range calculation needs. A third scheme, aggregating by sensor and day before forming the ratio, answers a different question, what a typical recorded day looked like for one vehicle, and is used only where that framing is the intended one.

### 2.5.4 Urban canyon model

To confront the measurements with theory, each point is given a geometry. A street is idealised as an infinitely long canyon: a flat roadway of width $W$, flanked by two continuous building rows of height $H$ (Figure 2.3). The ratio $H/W$ is the single parameter of the model [5]. Here $H$ is the mean height of the buildings within the 50 m buffer and $W$ the street width matched from OpenStreetMap, so the ratio is computed per point rather than averaged over a city.

The global horizontal irradiance the sensor should see under an open sky is the sum of the two components the model treats separately,

$$\mathrm{GHI} = \mathrm{BHI} + \mathrm{DHI} = \mathrm{DNI}\,\sin\gamma_s + \mathrm{DHI}, \tag{2.5}$$

with $\gamma_s$ the solar elevation. CAMS supplies the three terms separately, which is what allows a loss to be attributed to one component rather than to the total. Diffuse radiation is assumed isotropic, so what is lost depends only on how much of the sky dome remains visible, the sky view factor. With $\alpha$ the obstruction angle subtended by the facades,

$$L_{\mathrm{diffuse}} = 1 - \cos^2\alpha, \tag{2.6}$$

which depends neither on latitude, nor on season, nor on the orientation of the street. Direct radiation depends on all three: it is lost whenever the solar elevation falls below the obstruction angle in the direction of the sun, so its computation requires the declination, the hour angle and the street azimuth. The two components are then combined in proportion to the diffuse fraction of the incoming radiation.

The vehicle's position across the canyon matters. The obstruction angle differs between the middle of the street and its edge, and the resulting losses differ by roughly a factor of two at a given $H/W$. This maps onto the moving and parked distinction of Section 2.5.1, since a driving vehicle is in the middle and a parked one at the side. A GPS position does not resolve lateral placement within a street, so only the mid-street case is evaluated, and that factor of two remains an unresolved structural uncertainty of the comparison.

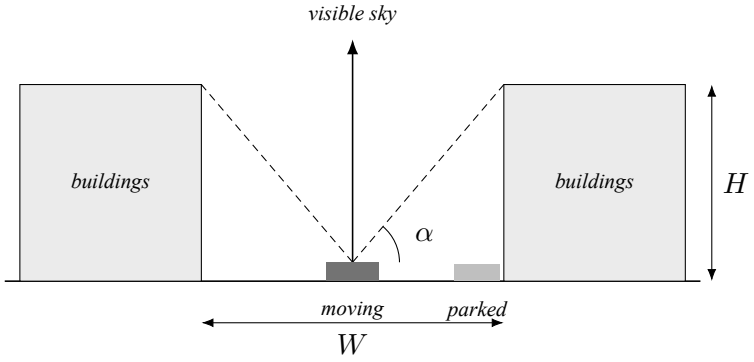


**Figure 2.3:** Idealised urban canyon. The street is treated as infinitely long, of width $W$, between two continuous building rows of height $H$. The obstruction angle $\alpha$, and with it the visible fraction of the sky dome, differs between the middle of the street and its side, which is why the moving and parked cases are not interchangeable.

## 2.6 Reproducibility and code organisation

Every script opens with a single configuration block, holding the paths, the campaign, the sensor, the active variant and the thresholds, and nothing configurable appears below it. Physics and reference

data live in shared modules rather than being restated: the land-cover nomenclatures, the building-height and street-width joins, and the canyon model of Section 2.5.4 each exist once and are imported by every script that needs them, so that a correction propagates instead of having to be remembered in three places.

Outputs carry their own provenance. Figures are written at 300 dpi into a directory named after the data variant that produced them, and each carries a footnote recording the campaign, the sensor and the variant tag.

# 3 Results and discussion

## 3.1 Sampling biases

The standard campaign rests on two sensors that did not record the same thing. J31 contributes about four times as many records as J30, over a much longer window: J30 stops on 25 October 2022 after four months of vehicle operation, J31 continues until 21 May 2023. J30 also stays inside Lisbon, where J31 covers Cascais, Sintra and the long trips. Any result computed on the pooled dataset is therefore mostly a J31 result, and any comparison that separates the two sensors separates two periods and two territories at the same time.

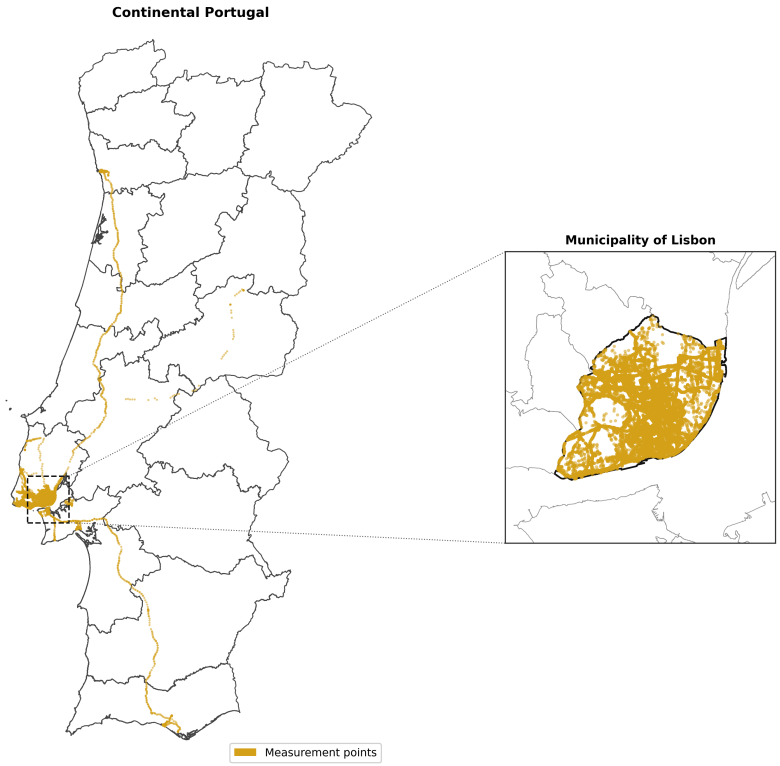


**Figure 3.1:** Spatial distribution of the standard-campaign measurements, `sd_cleaned` variant. The inset covers the Lisbon municipality, where the two blanks are the Monsanto forest park to the west and the airport to the north.

The records concentrate on Lisbon and Cascais, with a few long trips reaching Porto to the north and Faro to the south (Figure 3.1). Density is highest in the city centre and along the main axes. Urban environments are strongly over-represented, and Lisbon carries enough weight to drive any statistic computed on the unrestricted dataset. Two cautions attach to reading such a map. A dense area reflects the time spent there as much as the ground covered, so a bright zone may be a commuting axis or a single long-stay parking place. And only daylight records are retained, which makes this a map of

exploitable coverage rather than of where the vehicles went.

Broken down by land-cover class, the campaign is dominated by residential fabric, tourist facilities and the road network. The record count of a class is not a measure of how well that class is sampled. The 7,000 records of the maritime pine forest class come almost entirely from one site near Cascais, at a participant's home, and the most populated class of all, continuous horizontal residential fabric with 68,000 records, has its densest point at that same address. The tourist facilities class is concentrated on the Campo Grande campus. A handful of parking areas, the campus, Arroios, Benfica and above all Cascais, carry a large share of the records, each standing for weeks of minutes accumulated at one coordinate. Figure B.2 shows the per-class distribution behind these observations.

How much this matters is visible in what a geographic filter removes. Restricting the regional extent to the Lisbon municipality takes the dataset from 178,000 to 85,000 records, and the continuous horizontal residential class from 68,000 to 700, because the Cascais hot spot falls outside the boundary. The maritime pine forest, industry and commerce classes disappear from the sample altogether. The municipal and regional variants are not two views of the same population.

Temporal coverage is uneven in the same way. March carries no data at all, a complete absence consistent across every variant and matching a gap independently documented for J31, and January rests on two sensor-days, well below what any per-day normalisation requires. Monthly results presented later in this chapter have to be read against the number of sensor-days behind them rather than against the number of records.

## 3.2 First results of the tuk-tuk campaign

The tuk-tuk campaign is preliminary by construction: one vehicle, 24 days of exploitable data, and a radiometric calibration still at its default coefficients. No irradiance or shading result is reported from it here, since any such figure would be expressed in raw sensor units and would move once the calibration session is performed. The figures below use the `sd_cleaned` variant and sensor J33.

Of those 24 calendar days, 17 carry a non-zero travelled distance, for a total of 430 km, about 25 km per active day (Figure 3.2). The vehicle operates on most working days, in short repeated runs rather than long journeys. The operator's own estimate is 30 km to 50 km per day over six days a week. A day at zero distance is not necessarily a day without service, since the present outputs do not separate an idle day from a recording interruption.

Of the 6,282 recorded transitions, 61.6 % are classified as parked and 38.4 % as moving, with no records before 09:00 (Figure 3.3). The standard campaign, for comparison, is parked 81.1 % of the time. That contrast is the reason this campaign was launched: a tourist tuk-tuk drives through the middle of the day, when irradiance is highest and shading lowest, and it inverts the parking bias that dominates volunteer-based campaigns [5]. The direction of the difference is informative as no hourly bin reaches the reliability threshold.

The environment is homogeneous in a way the standard campaign never was. Only two COS2023 classes are populated, "continuous vertical residential fabric" and "road network", and the CLC rural

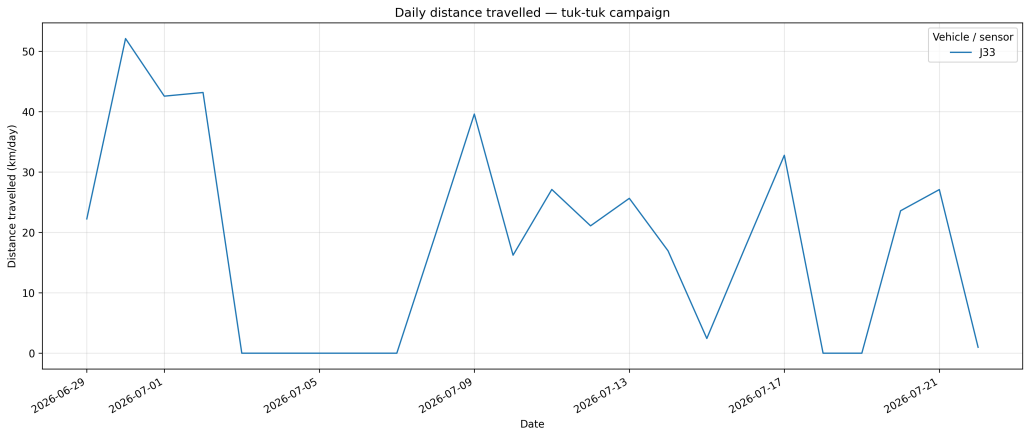


**Figure 3.2:** Distance travelled per calendar day, tuk-tuk campaign, `sd_cleaned` variant.

class is empty in every variant. The urban and rural contrast is not under-sampled here but structurally absent, so nothing can be said about it from this campaign. In exchange the dataset is a single context, dense historic centre, tall continuous facades and narrow streets, which is what makes it the more suitable of the two campaigns for a shading study once calibrated.

At 25 km per active day and a specific consumption between 10 kW h and 15 kW h per 100 km, the vehicle needs 2.5 kW h to 3.75 kW h of traction energy per day. One installed kilowatt-peak collects in Lisbon on the order of 1.5 kW h per day [3], and up to 2.3 kW h for a vehicle circulating rather than parked [2], so it would cover between roughly 40 % and 90 % of that demand. The irradiance term is transferred from published Lisbon values rather than measured on this vehicle, and the roof of a tuk-tuk cannot carry a full kilowatt-peak.

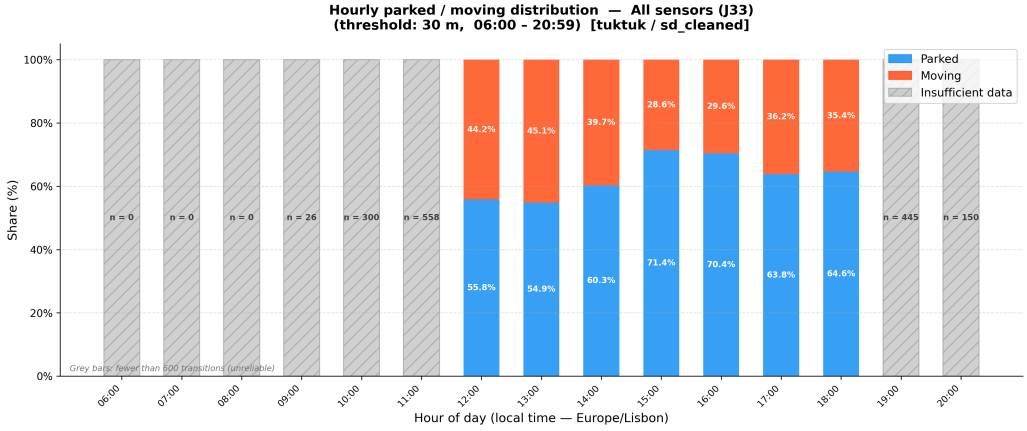


**Figure 3.3:** Hourly distribution of parked and moving records, tuk-tuk campaign, `sd_cleaned` variant.

## 3.3 Validation of the irradiance reference

Every loss reported in this chapter is a difference between what the vehicle measured and what CAMS says was available, so the second term has to be established first. Against the FCUL rooftop pyranometer, a fixed point with an open sky, the CAMS estimate gives regression slopes between 0.98

and 1.01 and coefficients of determination between 0.95 and 0.98. Against the on-board sensor the same comparison behaves quite differently: the coefficients fall to between 0.69 and 0.87, every slope exceeds unity, between 1.08 and 1.16, and the scatter separates into two populations, a dense cloud along the 1:1 diagonal and a near-vertical trail in which CAMS stays high while the sensor collapses (Figure 3.4).

The same estimate agrees with an unobstructed ground reference and separates from a moving sensor, so what changes between the two situations is not the estimate but the obstruction: the vertical trail is localised shading that a satellite pixel of several

Isolating built shading from cloud attenuation requires a subset of minutes in which the second is negligible, and the criterion adopted keeps those for which $\mathrm{GHI}/\mathrm{GHI}_{\mathrm{clear}} \geq 0.95$. The threshold is not arbitrary. Heliosat-4 computes the all-sky irradiance as the clear-sky value multiplied by a cloud index, and when the cloud-detection scheme classifies a pixel as cloud-free that index is exactly one [7], so the criterion selects the pixels the satellite itself declares free of cloud. Its limit lies in the same place: a field of scattered cumulus can be classified cloud-free once integrated over a pixel of three to seven kilometres, so the clear-sky subset still retains a fraction of minutes that were genuinely partly cloudy.

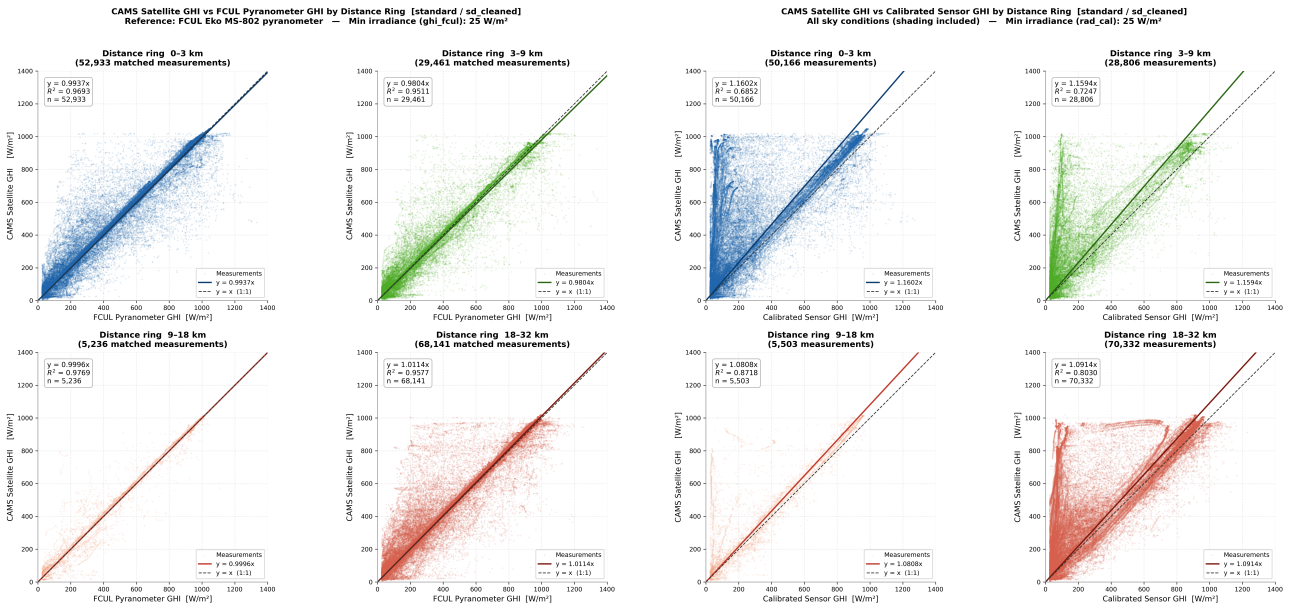


**Figure 3.4:** CAMS all-sky irradiance against a fixed ground reference (left, the FCUL rooftop pyranometer) and against the calibrated on-board sensor (right). The near-vertical trail on the right is localised shading, invisible to a kilometre-scale satellite pixel.

## 3.4 Vehicle behaviour

Whether a vehicle is driving or standing decides which of two regimes governs its solar yield: the conditions of a route, or the exposure of a parking place. The classification of Section 2.5.1 is applied here to the standard campaign, `sd_cleaned` variant.

Of 184,909 recorded transitions, 81.1 % are classified as parked and 18.9 % as moving. The parked fraction dips in the late morning, rises again through the early afternoon and reaches its maxima at the two ends of the day, where the samples also thin out (Figure 3.5). Month by month it runs from about a third in winter to above 90 % between June and September; March is empty. The vehicle is therefore stationary through the hours of highest irradiance, which places the yield of an on-board system first on where it is parked and only marginally on the conditions met while driving.

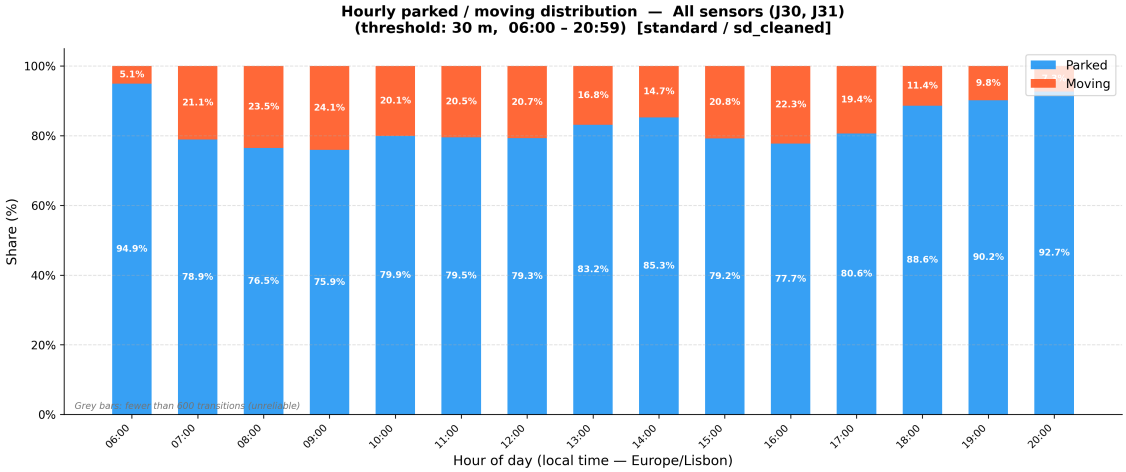


**Figure 3.5:** Hourly distribution of parked and moving transitions, standard campaign, `sd_cleaned` variant.

The same vehicles are almost never outside the city. The rural share of the standard campaign is 2.76 %, and the Portuguese classification says the same from the other side, with 95.1 % of the records in artificial surfaces. Hourly, that share is a shallow bell peaking around midday, from 0.53 % at 06:00 to 5.60 % at 14:00 (Figure 3.6); monthly it swings from 0.03 % in June to 9.50 % in April. What little rural remains is largely not rural: within the strict Lisbon boundary, 91.1 % of the points classified rural fall in a single class, transitional woodland-shrub, which at that location is the Monsanto forest park, an extended green space inside the city crossed by a road.

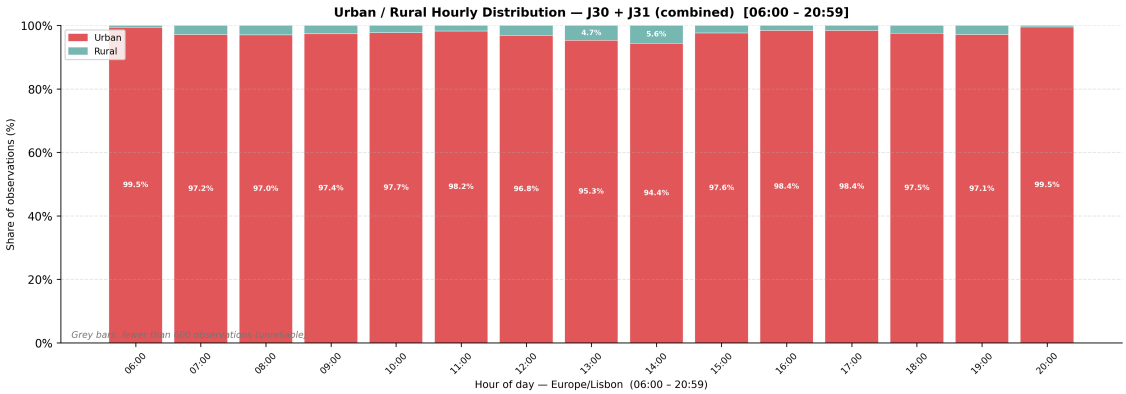


**Figure 3.6:** Hourly distribution of urban and rural records, standard campaign, `sd_cleaned` variant, sensors J30 and J31 combined.

Both monthly readings carry the same caveat, and it is the one already met in Section 3.1. The months in which the parked fraction is high are exactly the months in which J30 contributes, and J30

is a near-stationary sensor, around 95 % parked throughout its window; the rural share moves with the same composition.

## 3.5 Influence of land-cover class

Previous work measure just one axis: urban against rural [3–5]. Section 3.4 has just shown that axis to be unusable here. This section identifies what structures the losses instead, on the `hq` variant.

### 3.5.1 What structures the loss: sky openness

Computed class by class rather than along the urban/rural dichotomy, the relative autonomy losses range from 12.3 % to 70.9 % across the ten CLC classes that pass the reliability threshold, each formed by Equation (2.4) (Figure 3.7). The lowest belong to land under agriculture with significant natural vegetation (12.3 %), complex cultivation patterns (13.8 %) and the road and rail network (20.9 %); continuous urban fabric sits at 48.4 %; and the highest, at 70.9 %, belongs to transitional woodland-shrub.

Between those extremes sit the two classes that actually carry the campaign. Continuous and discontinuous urban fabric together hold the great majority of the records, an order of magnitude more than any other class above the threshold, and they separate cleanly: 48.4 % against 31.6 %, a gap of 17 points between two classes that a coarser reading would merge into a single urban category. The difference between them is density of built fabric and it is the largest well-sampled contrast the land-cover axis produces.

That ordering does not follow the urban and rural line. The classes that lose least are those with an open sky, whether urban or rural, and the class that loses most is rural: a dense canopy occludes as effectively as a canyon of facades, more effectively here. What the land-cover class captures is not the character of the environment but how much of the sky dome remains visible from the vehicle.

Two limits apply. Several sparsely populated classes report captured fractions above 100 %, physically possible on a curved roof but meaningless at those sample sizes. And the parked fraction varies strongly from class to class, so a class-level loss mixes the effect of the place with the effect of what the vehicle was doing there; the two are separated in Section 3.8.

### 3.5.2 What a finer classification adds

The two classifications are not two versions of the same product (Table 3.1). Two of the distinctions the Portuguese one draws have no CLC equivalent, and one of them matters here: continuous residential fabric is split by the dominant height of its buildings. For a study of a vehicle that stands still four fifths of the time under facades of varying height, that is not a detail.

That extra resolution produces the result of this subsection. The Portuguese classification splits what CLC records as continuous urban fabric into two morphologies (Figure 3.8). Predominantly horizontal residential fabric, 67,736 records, loses 32.96 %; predominantly vertical fabric, 30,125 records, loses 47.17 %. Fourteen points separate two classes that differ only in building height.

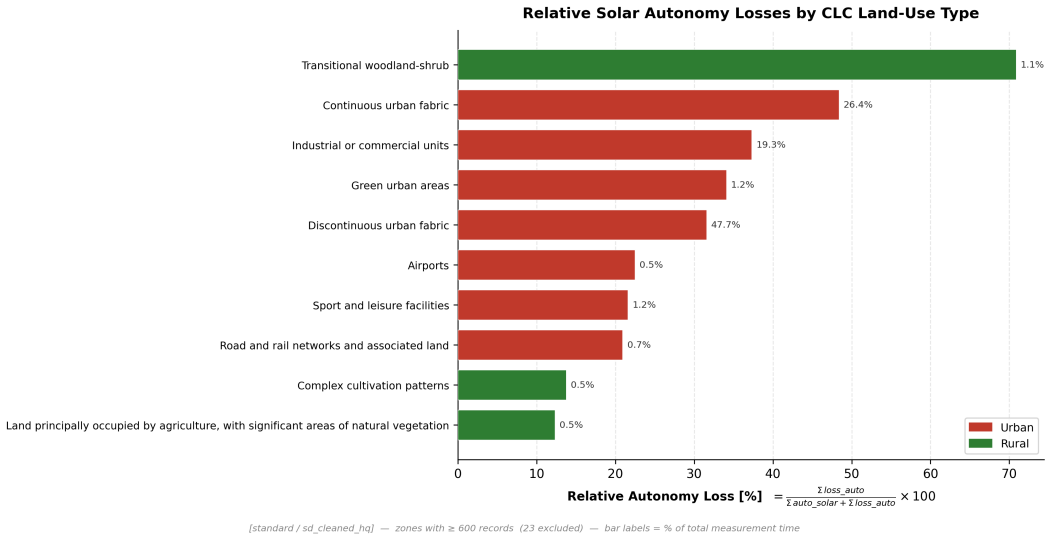


**Figure 3.7:** Relative autonomy loss by CORINE Land Cover class, standard campaign, hq variant. Only classes above the reliability threshold are shown.

**Table 3.1:** Structure of the two land-cover classifications used in this work.

| | CLC | COS2023 (Series 2) |
|---|---|---|
| Producer | EEA / Copernicus | DGT, Portugal |
| Hierarchical levels | 3 | 4 |
| Classes at the finest level | 44 | 93 |
| of which artificial surfaces | 11 | 37 |
| Minimum mapping unit | 25 ha | 1 ha |
| Minimum polygon width | 100 m | 20 m |
| Encodes building height | no | yes |

The reservation is one of resolution, not of method. Both classifications remain descriptions of a polygon, assigning one label to every point inside it, and neither can express that a vehicle stood on the sunlit side of a street rather than the shaded one. What does the work is the height and continuity of the buildings, not the nomenclature, which is the argument for measuring the geometry directly as Section 3.7 does.

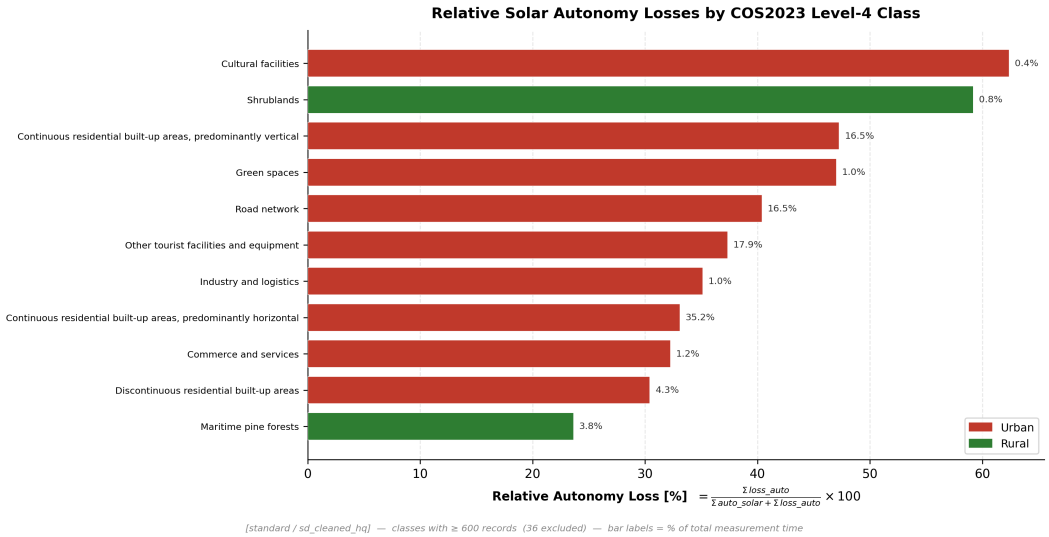


**Figure 3.8:** Relative autonomy loss by COS2023 level-4 class, standard campaign, `hq` variant.

## 3.6 Diffuse fraction

Diffuse radiation, assumed isotropic, reaches a vehicle from the whole visible sky rather than from one direction, and is therefore less easily intercepted by a facade than the direct beam. That argument underlies the canyon model of Section 2.5.4 and suggests that shading losses should weigh less heavily when the sky is diffuse. This section tests that expectation, on the `sd_cleaned_hq` variant.

The diffuse fraction $k_d = \mathrm{DHI}/\mathrm{GHI}$ is derived from the CAMS components. Plotted against the clearness index $k_t = \mathrm{GHI}/\mathrm{GHI}_{\mathrm{extra}}$ it traces a tight monotonic decrease (Figure 3.10): $k_d$ close to unity at low $k_t$, where the beam is negligible, falling to about 0.1 at $k_t \approx 0.8$. That is the shape the decomposition literature describes, so the split between diffuse and direct in the satellite product can be used to sort the measurements by sky condition. It is not, however, a measurement made on the vehicle: it is a model output for an area, over a flat unobstructed surface, and it contains neither the reflections from facades that a sensor in a street receives nor any effect of the obstruction itself.

The skies the campaign sampled are markedly bimodal (Figure 3.9). A single bin, around $k_d \approx 0.1$ to 0.15, holds 30.6 % of all records, a flat valley follows between 0.2 and 0.9, and the count rises again close to unity. Sorted into regimes, 44.7 % of the records fall under a clear sky and 28.7 % under an overcast one, which leaves the intermediate skies the least represented of the three.

Two modes, not a spread around a mean: the sky over Lisbon is most often either clear or covered, and the transition between the two states is faster than the time the vehicle spends in it. The two modes

are the two ends of the same relation, and they are visible again in Figure 3.10, where the point cloud follows the familiar $k_d(k_t)$ curve, dense near $k_t \approx 0.7$ with $k_d \approx 0.15$, dense again near $k_t \approx 0.2$ with $k_d$ close to one, and thin in between. The bimodality is therefore a property of the sky the campaign sampled.

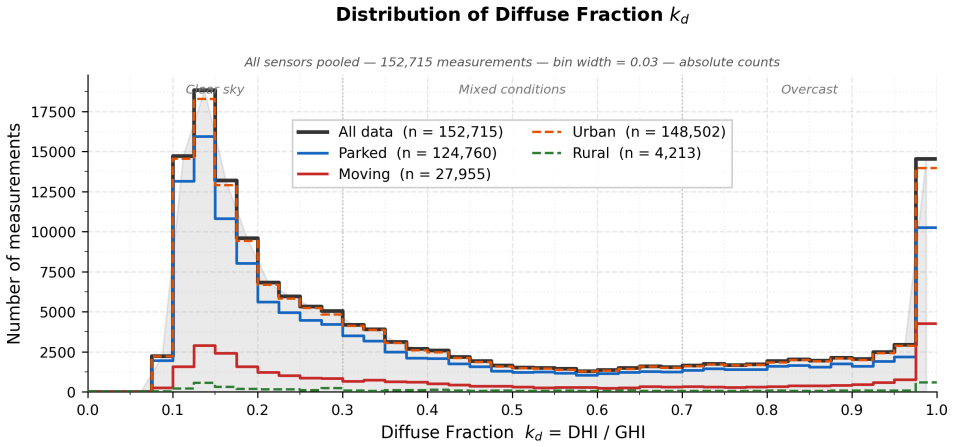


**Figure 3.9:** Distribution of the diffuse fraction $k_d$, standard campaign, `hq` variant, all sensors pooled. The clear-sky mode near $k_d = 0.14$ and the overcast mode against $k_d = 1$ are separated by a valley that holds a small minority of the records.

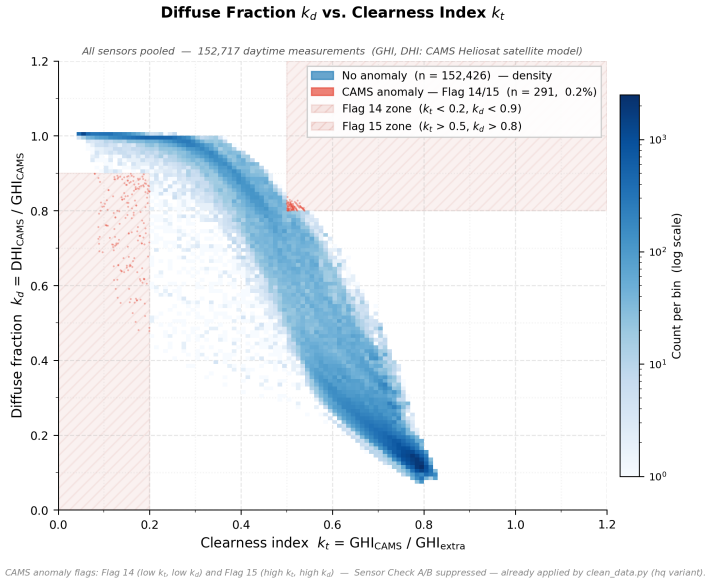


**Figure 3.10:** Diffuse fraction against clearness index, standard campaign, `hq` variant.

Sorted by diffuse fraction, the relative loss does not decrease as the sky turns diffuse. It traces a bell (Figure 3.11), rising to a maximum in the bin centred on $k_d = 0.35$ and falling away on either side: 53.5 % over the full dataset, 57.8 % restricted to Lisbon, 64.6 % on the most filtered variant. The maximum sits in the same bin in all nine variants examined, whatever the quality, clear-sky or geographic filter applied.

The expectation is not wrong, it is incomplete, because it describes only one of the two components. As the sky turns diffuse the part of the radiation a facade intercepts easily, the beam, does shrink,

and with it the mechanism the canyon model represents; but the total available radiation shrinks at the same time and faster. The relative loss is a ratio between the two, so it is maximal not where shading is worst but where the two trends cross, in the intermediate regime. The absolute loss, which is not a ratio, peaks instead in the bin at $k_d = 0.15$ and decreases from there, since there is less irradiance to lose once cloud sets in. The two maxima fall under different skies, and quoting either without the other is misleading.

Two reservations attach to this reading. Taken as medians rather than means, the relative loss jumps from about 12 % to about 68 % in a single bin step, so within a bin the distribution is not unimodal. And the regime in which the relative loss is greatest is the one the campaign sampled least.

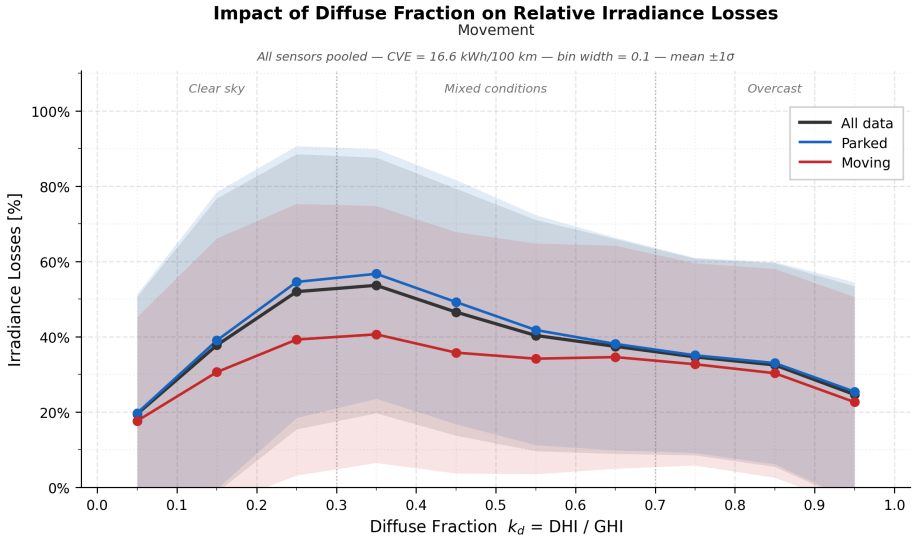


**Figure 3.11:** Relative loss by diffuse-fraction bin, mean and standard deviation, standard campaign, `hq` variant.

## 3.7 Urban geometry and shading

Section 3.5 concluded that what a land-cover class captures is the morphology of the built fabric. This section measures that morphology directly, first through building height alone and then through the height-to-width ratio of the street, on the `hq` variant.

Binned by the mean height of the buildings within 50 m, the relative loss rises from 30 % to 41 % in the lowest bin to 58 % to 69 % above 30 m, a factor of 1.7 to 2.0 across the range (Figure 3.12). The tallest bin is the most penalising, so height alone is a usable predictor and the direction of the relation is robust to every filter applied.

Adding the street width matched from OpenStreetMap gives a per-point $H/W$ ratio and allows the measurement to be placed against the theoretical loss of the canyon model (Figure 3.13). Requiring both a valid height and a valid width is costly: the sample falls to 98,355 points, 63 % of the height-only subset, and becomes confined to the Lisbon area. The comparison is one of shape, since the theoretical curve models the isotropic diffuse component only, at mid-street and for an assumed constant width,

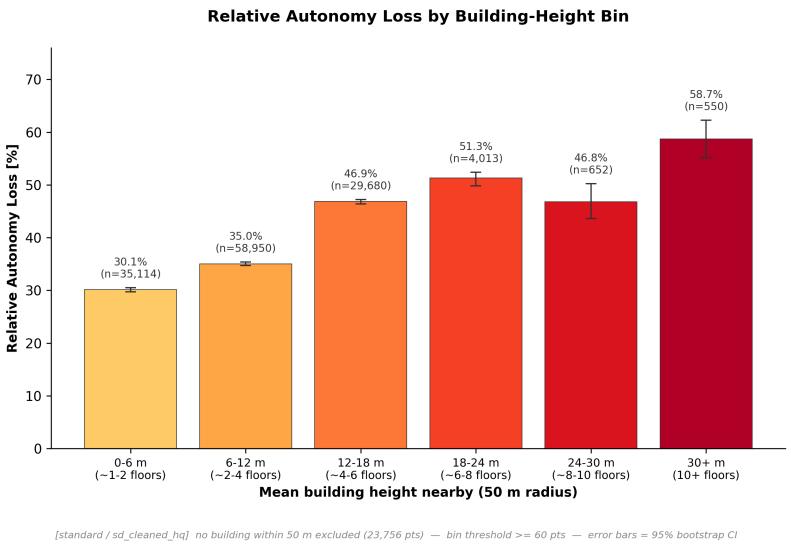


**Figure 3.12:** Relative loss by mean surrounding building height, standard campaign, hq variant.

whereas the measurement aggregates diffuse and direct under a real sky. Read that way, the two agree over most of the range and diverge where the measured relation ceases to be monotonic.

Part of that divergence has an identified origin. Stratified by clear-sky index, the clear and the overcast regimes each follow a significant relation between theoretical and measured loss, with $R^2 = 0.84$ and 0.73, while the intermediate stratum is erratic and not significant: the same regime in which Section 3.6 located the maximum of the relative loss. One structural uncertainty remains unresolved. At a given $H/W$, the middle of the street and its side differ by roughly a factor of two, and a GPS position does not say which of the two applies.

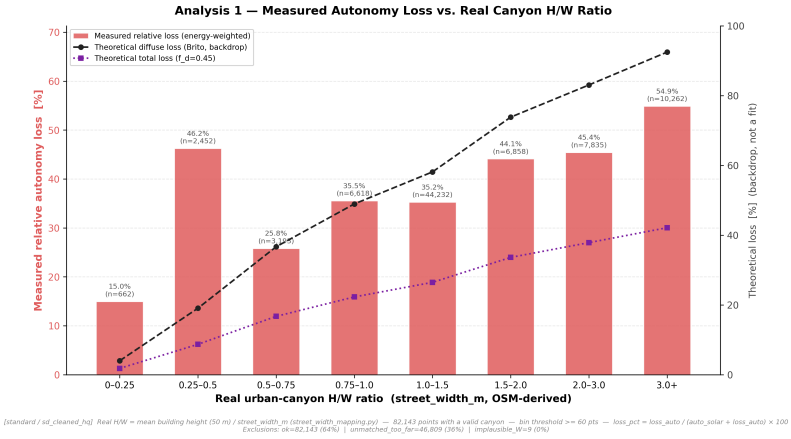


**Figure 3.13:** Measured relative loss against the real height-to-width ratio, with the theoretical diffuse-loss curve as a backdrop. Standard campaign, hq variant.

Three limits condition the above. The first is coverage: 15.4 % of the standard campaign has no building within 50 m, rising to 21.0 % inside Lisbon and 27.4 % for the tuk-tuk. That absence is not random and is largely legitimate, concentrating in genuinely open classes, from construction sites at 98 % to the road network at 47 %, while dense residential fabric stays at 1 % to 2 %.

The second concerns the detection method rather than the data. The building footprints of GlobalBuildingAtlas come from three sources; those derived from OpenStreetMap represent individual buildings, while the other two sometimes merge several buildings, or an entire block, into a single polygon. The heights themselves remain sound, but this work detects buildings by their centroids, so when a polygon stands for a block its centroid can fall well away from the building that actually obstructs the sky, and escape the 50 m radius entirely. Centroid detection is therefore reliable in OSM-sourced neighbourhoods and unreliable elsewhere, without this being visible in any coverage statistic; interactive inspection of the joined points confirms the contrast between the two kinds of district.

The third is of the same family. The street-width estimate settles near 6 m in every land-cover class, which is a selection effect rather than a result: the OSM tags are only consulted where a building height exists, that is where buildings are close, that is where streets are narrow. The value is consistent for those streets and underestimates the city-wide average. The second and third limits share a remedy, detecting buildings as polygons rather than as centroids, which is the first item of the work proposed in Chapter 4.

## 3.8 Shading losses

This section brings the preceding ones together into the quantity the work set out to estimate: how much driving range the shading of the urban environment costs, and under which conditions.

Expressed per unit of installed peak power, the range the collected energy would provide follows the season closely (Figure 3.14). The achieved range rises from 9.0 km/(day kWp) in February to 22.9 km/(day kWp) in May, then falls to 4.9 km/(day kWp) in December, for an annual mean of 14.8 km/(day kWp) against a clear-sky potential of 23.5 km/(day kWp). The loss follows the same shape rather than compensating it, at 4.5 km/(day kWp) in February, 12.9 km/(day kWp) in August and 3.6 km/(day kWp) in December: shading removes a share of what is available rather than a fixed amount, so the months that offer the most also lose the most. That share is close to constant from February to September, between 33 % and 37 %, and rises to 42 % to 46 %. The monthly reading remains subject to the coverage constraint of Section 3.1.

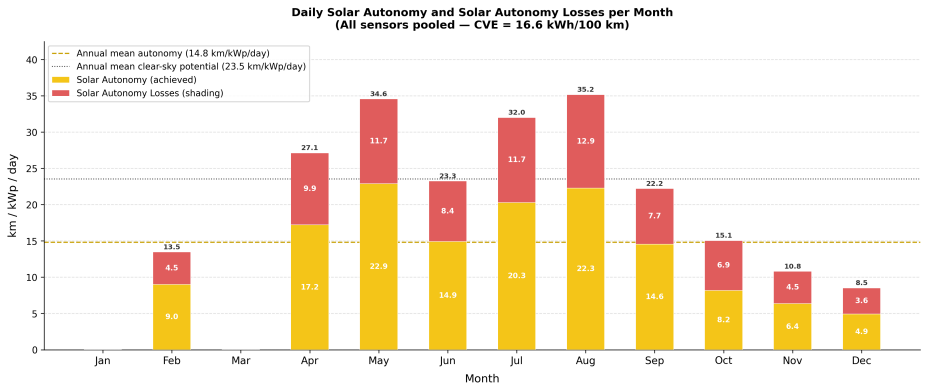


**Figure 3.14:** Monthly solar range and shading loss per unit of installed peak power, standard campaign, `hq` variant.

The expectation on movement was that a moving vehicle would lose less than a parked one: it meets a diversity of conditions instead of one, and the middle of a street is a more open position than its side. That is what the measurements show, and it is the most robust result of this work. Across the seven standard variants, the relative loss is lower while moving than while parked in 91 % of the comparable hourly and monthly bins, by 10 to 17 percentage points on average depending on the variant (Figure 3.15). The exceptions are not scattered: eleven of the thirteen fall either in February or at 10:00, the two bins where the two populations are closest, and the gap there stays below one point in most variants. The ordering survives every quality, clear-sky and geographic filter; it holds within the urban class taken alone, where moving stays about 10 points below parked in the broadest variants and 16 in those restricted to Lisbon; and it holds, directionally, on the single reliable monthly bin of the tuk-tuk campaign.

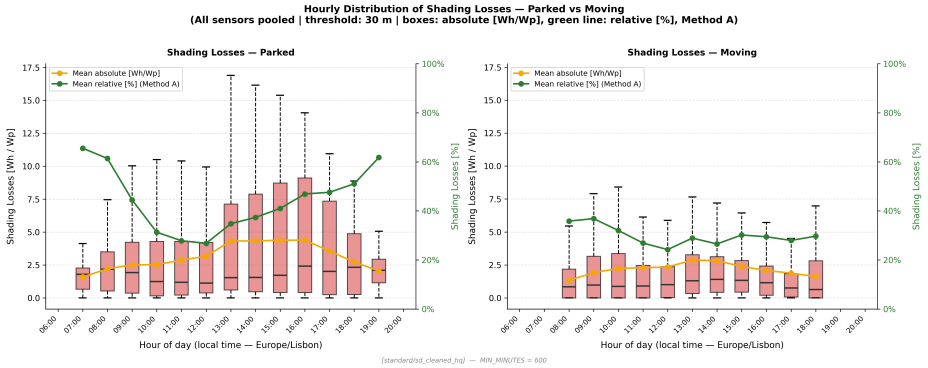


**Figure 3.15:** Shading loss by hour, parked against moving, standard campaign, `hq` variant. Boxes give the absolute loss, the line the relative loss formed by Equation (2.4).

The symmetric expectation, that fewer and lower buildings in rural surroundings should mean smaller losses, cannot be tested here. Once the land-cover join was corrected, the rural class lost the recurring misclassified location that had been supplying most of its volume, and what remains is too rare to fill a comparison: no more than two of fifteen hourly bins, or four of twelve monthly bins, reach the reliability threshold in the most permissive variant, and none does in the most filtered ones. The four-way crossing of land use with movement status, one of the intended outputs, produces no reliable rural moving bin in any variant, so the crossing cannot be formed and is not reported. Of the fifteen bins where both classes are jointly reliable, eleven place the relative loss higher in the rural class, the opposite of the expected ordering, which is consistent with Section 3.5.1 where the class that lost most was rural under tree canopy. Those bins are too few to support a conclusion.

# 4 Conclusions and future work

## 4.1 What this work produced

The internship set out to estimate the solar potential of a vehicle in a city, and the loss of the campaign data made that estimate impossible. What it produced instead is the chain that will allow it to be made, from the raw sensor export to the interpreted figure, held together by a naming convention that makes every file describe the treatment it has received (Chapter C). It also launched a second measurement campaign and brought its data into that chain.

The results divide into those that hold and those that cannot yet be tested. The movement status of the vehicle proved the robust axis: relative losses are lower while driving than while parked in 91 % of the comparable bins, under every filter, and within the urban class taken alone. Building height is a usable predictor of loss, with a factor of two between the lowest and the tallest surroundings. Conversely, the urban and rural contrast that organised the earlier work is not testable on this dataset, and the class-level analysis suggests it was a proxy for the openness of the sky.

One observation follows from Section 3.4: since these vehicles spend most of their time parked, a solar canopy over a parking area is the same technology relieved of that constraint, and it is not obvious from the evidence gathered here that per-vehicle photovoltaics outperforms it.

## 4.2 A proposed method: polygons instead of centroids

The limitation identified in Section 3.7 has a direct remedy. Detecting buildings by their centroids fails wherever a footprint stands for a block rather than for a building; using the polygons themselves removes the failure at its source, which means taking the GlobalBuildingAtlas data in its published polygon and LoD1 form. A first implementation exists but has not been used here.

The same geometry allows the shading model to be rebuilt around the distinction the present one only approximates, by treating the two components of the radiation separately (Figure 4.1). For the *direct* component, a segment traced from the GPS position towards the sun either intersects the volume of a building or it does not: the beam contribution is lost entirely or received in full. That binary quantity matches the physical mechanism better than a continuous attenuation, and would explain the abrupt transition observed in Section 3.6. For the *diffuse* component, assumed isotropic, what matters is the fraction of the sky dome still visible once the surrounding volumes are subtracted, computed from the same polygons.

Validating it requires a separate measurement of the direct and diffuse components on the vehicle, which the present instrumentation cannot provide.

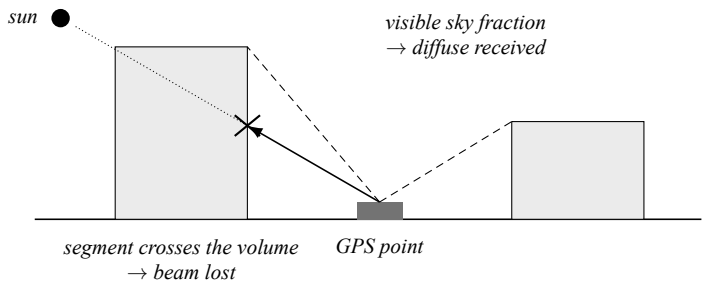


**Figure 4.1:** Proposed treatment of the two radiation components. The direct beam is either intercepted by a building volume or received in full; the diffuse component is scaled by the fraction of the sky dome that remains visible from the same point.

## 4.3 Continuing the tuk-tuk campaign

The immediate step is a calibration session against the FCUL reference pyranometer, to replace the placeholder coefficients that make the calibrated column a copy of the raw one. Beyond that, the campaign needs to continue: 24 days of one vehicle established a usage profile, where a season of several would give the shading figures the sample the standard campaign never had here, homogeneous and free of parking bias. The chain is already in place, so each additional vehicle costs almost nothing, and the order-of-magnitude estimate of Section 3.2 would become a measurement rather than a transfer from published values.

## 4.4 What should be re-examined

Several choices made here were reasonable at the time and deserve to be revisited rather than inherited.The 50 m buffer used for building height, the three-kilometre threshold below which a sensor-day is treated as non-vehicular, and the clear-sky criterion share the same status, defensible but never tested for their influence on the conclusions. Aggregating by hour of the day belongs on the same list, since it compares 14:00 in December with 14:00 in June; aggregating by solar elevation band would separate the daily pattern from the seasonal one.

One question remains open: which land-cover classification should serve as the primary one. The finer Portuguese product resolves what matters here, the European one keeps the results directly comparable beyond Portugal.

The scripts, the mapping modules and the analysis notes are available at …

# 5 Skills assessment

Of the four competencies defined, three were retained at the start of this internship as those the work would call on: elaborating and transforming materials C1, characterising materials and multiscale devices C2, and producing and managing energy C3. The associated learning outcomes were set out with my supervisor at the outset, and what follows is my own assessment of where I stand against them.

The first outcome was to analyse solar radiation and photovoltaic processes in a VIPV context. The central question, how much of the available resource actually reaches a vehicle in a city, cannot be answered by running a model; it requires deciding what the model represents. I had to connect the physical mechanisms at work to the assumptions, inputs and outputs of the numerical treatment rather than take the computation as given: why diffuse radiation, assumed isotropic, depends only on the visible fraction of the sky dome while direct radiation depends on latitude, season and street orientation (Section 2.5.4); why truncating the loss at zero is a deliberate asymmetry and not a numerical convenience; why a street width used in place of a facade-to-facade distance makes the resulting loss a conservative upper bound rather than an estimate.

Asking what a quantity means before asking what value it takes is what I take away from this competency. It is also what allowed me to establish, in the course of the work, that the saturation of one index at unity was not a model artefact, as had been assumed, but the structure of the method itself (Section 3.3).

Three outcomes were defined here: processing and cleaning experimental datasets, applying suitable statistical and numerical tools, and critically validating model predictions.

The data are multiscale by construction: measurements at the minute, a satellite product resolved over kilometres, land cover mapped at one hectare or at twenty-five, building heights joined within a fifty-metre radius. Making these coexist in a single row of a table was a substantial part of the work, and it taught me to check formats, units, missing values and intermediate outputs systematically rather than trust that a script which runs has run correctly. Python was the main tool throughout, and I moved from isolated scripts to a structured chain in which each stage appends to its input without overwriting it and a filename records the exact sequence of treatments applied (Section 2.1). Designing something reproducible, rather than obtaining a result once, is the more durable skill.

Critical validation is where this internship taught me most. A prediction was never accepted because the code executed: the reference irradiance was checked against an independent ground measurement before being used as a reference at all (Section 3.3), and the sample itself was audited before any loss figure was read from it (Section 3.1). That audit revealed that an apparently well-populated land-cover class could rest on a single location, and that a distance-based comparison was in fact a comparison between two vehicles. Recognising that a result can be numerically correct and still

answer a different question than the one asked is the most useful thing I learned.

The remaining outcomes concerned energy generation under non-standard conditions, the evaluation of VIPV yield, and the communication of results. Evaluating photovoltaic generation on a curved surface, in motion, partially and intermittently shaded meant expressing it in terms an energy assessment can use, collected energy per unit of installed peak power, loss relative to an unobstructed reference, driving range gained per day, and stating the constants involved explicitly, since two studies cannot be compared if their assumed consumption differs.

Communication proved a substantial part of the internship. I had to present progress to people with different areas of expertise, and separate a physical explanation from a methodological choice or an implementation detail. Attending the IEA PVPS Task 17 meeting as an observer showed me how the same results are framed differently for a research and for an industrial audience. Working throughout in English, and preparing figures (and code meant) to be read by others rather than by myself, made clear that a result becomes useful only once its method and its limits can be stated plainly.

The internship gave me a concrete experience of the iterative character of research: cycles of reading, method definition, implementation and verification, several of which ended in a correction rather than a result. Losing the greater part of the campaign data also required reorienting the objective, from producing a figure to building the chain that would allow one to be produced, a lesson in adapting the aim of a project to what its material permits. Working in a group whose members had their own projects also required judging when a difficulty deserved a question and when it deserved another attempt.

# 6 Ecological transition and sustainable development

The dominant source of greenhouse-gas emissions attributable to this internship is a single item: the return journey between Nantes and Lisbon, made by air, on a direct flight in each direction. The institutional travel declaration returns 402 kg of $CO_2$-equivalent for 2,257 km travelled, an intensity of 178 g/km (Figure 6.1).

Every journey made during the stay itself, daily commuting to the campus and travel within the country, was made by metro, bus or train, and contributes a negligible share of that total. Beyond travel, the internship generated no specific pollution: the work was computational, and its own footprint is indistinguishable from the ordinary running of a university office.

**Figure 6.1:** Travel declaration for the internship: 402 kg $CO_2$eq over 2,257 km, on a direct flight in each direction.

Portugal has gone a long way in decarbonising its electricity. In 2024 renewable sources covered 70 % of electricity consumption, distributed as shown in Table 6.1 [14].

**Table 6.1:** Electricity supply of mainland Portugal in 2024.

| Renewable, 70 % | | Other sources | |
|---|---|---|---|
| Hydro | 28 % | Natural gas | 10 % |
| Wind | 27 % | Net imports | 20 % |
| Solar | 9 % | | |
| Biomass | 6 % | | |

That headline figure should not be mistaken for a decarbonised energy system. Renewables accounted for 34.7 % of gross final energy consumption in 2022, against 61 % of electricity generation the same year [15]. The gap between those two numbers is the part of the transition that remains to be done, and it lies almost entirely in transport, buildings and industrial heat, which is also, precisely, where the subject of this internship sits.

The country is meanwhile exposed to the physical consequences of warming: 2024 was its fourth warmest year since 1931 and brought eight heatwaves [16], which in Lisbon means pressure on water management, on the thermal adaptation of buildings, on urban greening and on collective mobility.

CIÊNCIAS ULisboa is a teaching and research faculty on the Campo Grande campus. Laboratories, scientific instruments and computing facilities give it an energy and resource consumption well above that of ordinary offices, and its sustainability policy addresses campus management, teaching, research and outreach together [17].

Its Living Laboratory for Sustainability, established in 2015, tests such solutions on the campus itself. Two of its projects are Solar Campus and the Solar Urban Garden, the latter combining photovoltaic generation with cultivation on the same ground, while others address biodiversity, active mobility and air quality [18]. The faculty reports on all of it in a sustainability report covering 2022–2024, built on 115 indicators of which 20 are key and mapped onto the United Nations Sustainable Development Goals [19].

The subject of the internship belongs to the gap identified above. Vehicle-integrated photovoltaics generates electricity on board, reducing both the energy drawn from the grid and the network reinforcement that mass electrification would require, a contribution to exactly the sector in which Portugal's renewable share is lowest.

That contribution depends on how much of the available resource actually reaches the vehicle. The losses imposed by the urban environment are large and vary considerably from one setting to another (Section 3.5). Characterising urban shading is therefore part of the environmental assessment of the technology.

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# Appendix A

# Data sources

Table A.1 lists the external datasets combined with the on-board irradiance measurements, with the native resolution of each product, not the resolution at which it is used after the joins described in the Methods chapter.

**Table A.1:** External datasets used in this work.

| Dataset | Provider | Native resolution | Vintage | Use here |
|---|---|---|---|---|
| CAMS Radiation Service (Heliosat-4 [7] / McClear [20]) | Copernicus Atmosphere Monitoring Service | Cloud properties from MSG/APOLLO every 15 min, 3 km at nadir (3 km to 7 km over Lisbon); aerosol, water-vapour and ozone inputs every 3 h at 0.4° | continuous | Reference GHI, clear-sky GHI and extraterrestrial irradiance, plus BHI, DHI and DNI. Extracted at 1 min on a 0.05° grid |
| `pvlib` | open-source community | — | v — | Solar elevation and azimuth at each measurement point |
| CORINE Land Cover | Copernicus Land Monitoring Service / EEA | 100 m; minimum mapping unit 25 ha; minimum width of linear features 100 m | 2018 | Secondary land-cover classification, retained for European comparability; legend in Chapter D |
| COS2023, Carta de Uso e Ocupação do Solo, Série 2 | DGT (Direção-Geral do Território), Portugal | Vector; minimum mapping unit 1 ha, minimum polygon width and minimum distance between lines 20 m | 2023 | Primary land-cover classification; the only nomenclature of the two that encodes building verticality; legend in Chapter E |
| GlobalBuilding-Atlas (GBA.Height, GBA.LoD1) | Technical University of Munich | Height raster at 3 m; RMSE 4.1 m in Europe; height completeness above 97 % | PlanetScope imagery, 2019 | Building height $H$, through footprint centroids compiled by U-Shift/CERIS |

*continued on next page*

*Table A.1 continued from previous page.*

| Dataset | Provider | Native resolution | Vintage | Use here |
|---|---|---|---|---|
| OpenStreetMap | OSM contributors | Vector, no fixed resolution; width from an explicit `width` tag, else lane count $\times$ standard lane width, else a per-road-class default | continuous | Street geometry and street width $W$ for the urban-canyon ratio $H/W$ |

# Appendix B

# Supplementary figures

## B.1 Dataset characterization

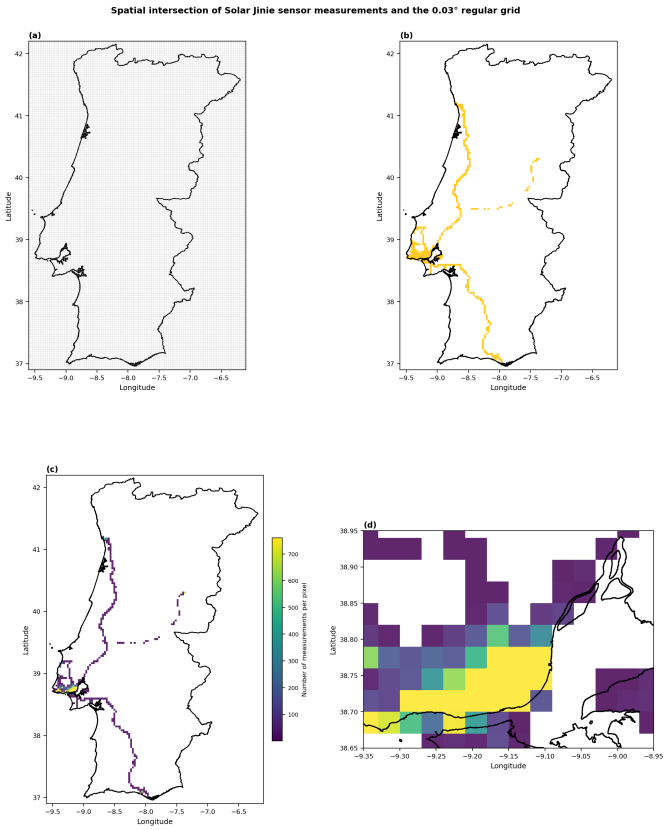


**Figure B.1:** Measurement coverage of the standard campaign aggregated on a 0.03° grid (about 3 km at Portuguese latitudes): reference grid, occupied pixels, density over the national extent, and close-up on the Lisbon municipality. Daylight records only. The colour scale is capped at the 95th percentile of this dataset and is not comparable with the equivalent figure for another campaign.

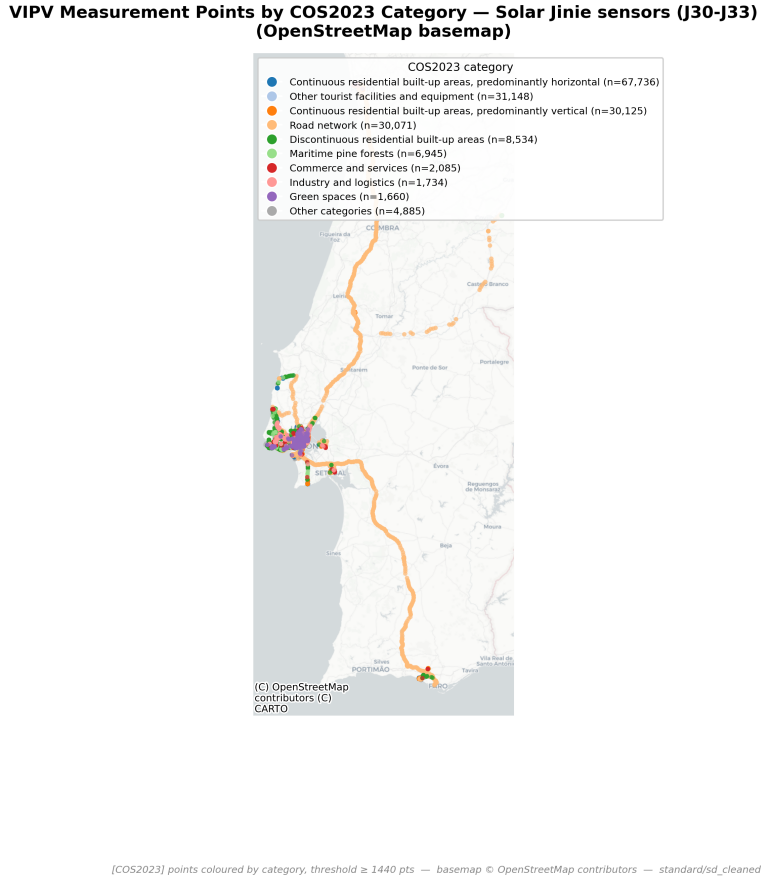


**Figure B.2:** Spatial distribution of the standard-campaign measurements by COS2023 class. Several classes rest on a small number of locations, most visibly the maritime pine forest class, concentrated on a single site near Cascais.

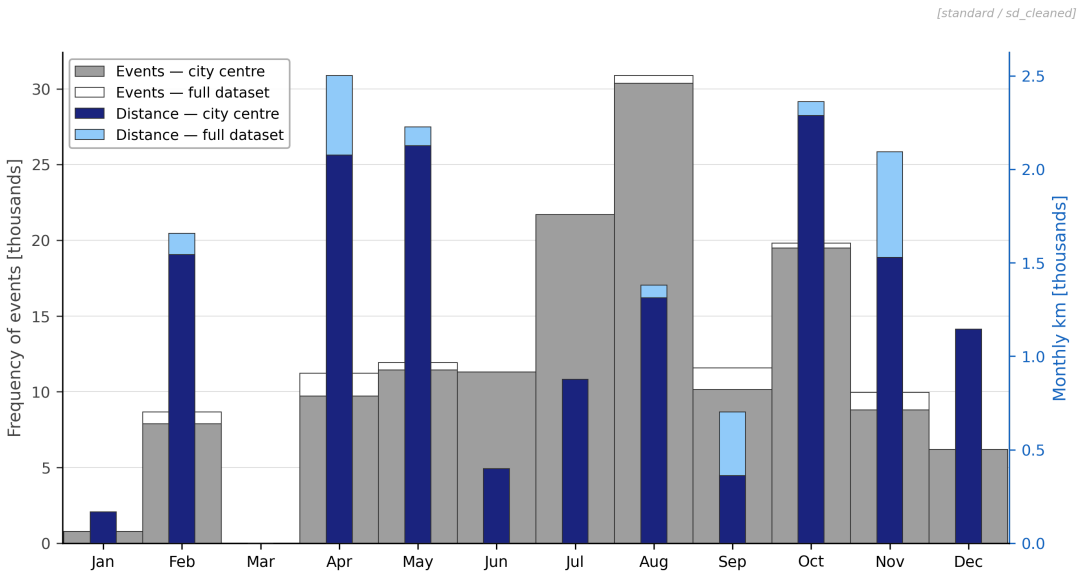


**Figure B.3:** Monthly record counts and travelled distance, standard campaign, `sd_cleaned` variant. March is absent and January rests on two sensor-days, which is what Section 3.1 reports.

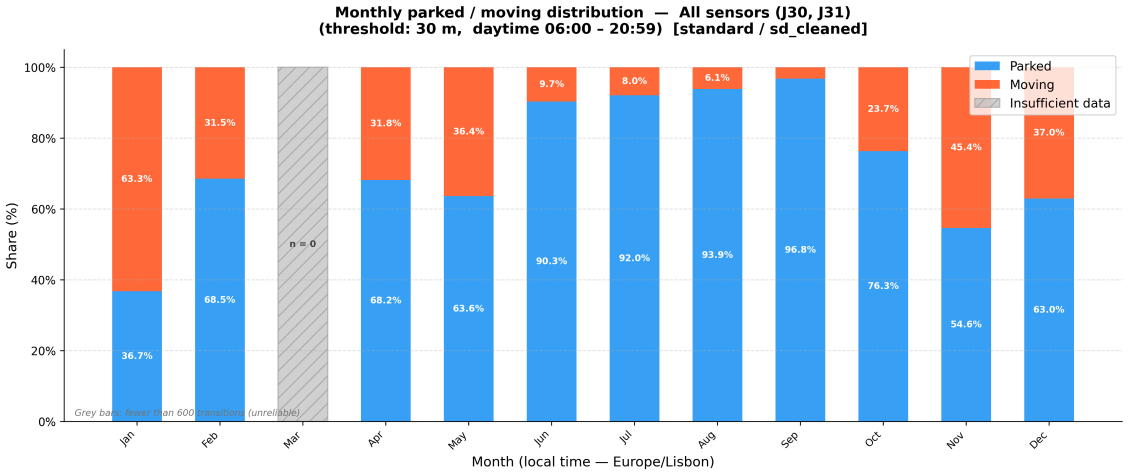


**Figure B.4:** Monthly distribution of parked and moving transitions, standard campaign, `sd_cleaned` variant. The summer rise coincides with the arrival of the near-stationary sensor J30 in the sample and is not a seasonal effect (Section 3.4).

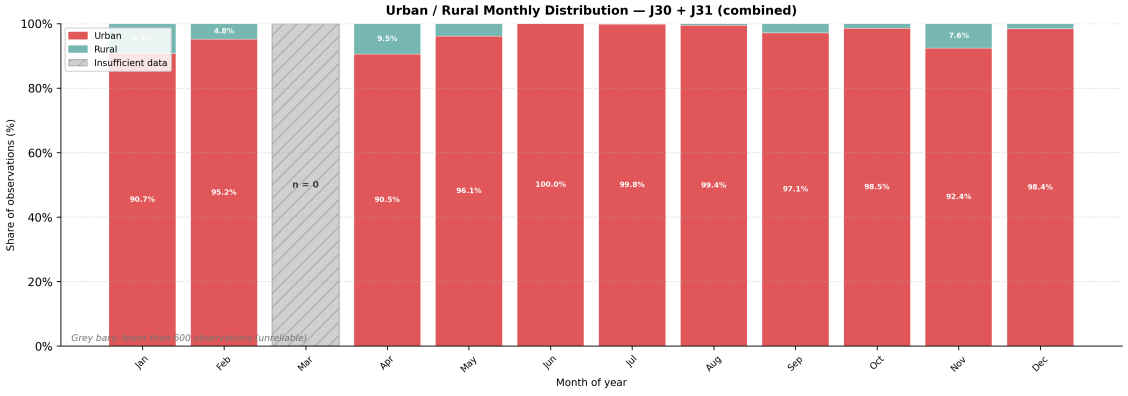


**Figure B.5:** Monthly distribution of urban and rural records, standard campaign, `sd_cleaned` variant. The swing follows the same change in sensor composition as the figure above.

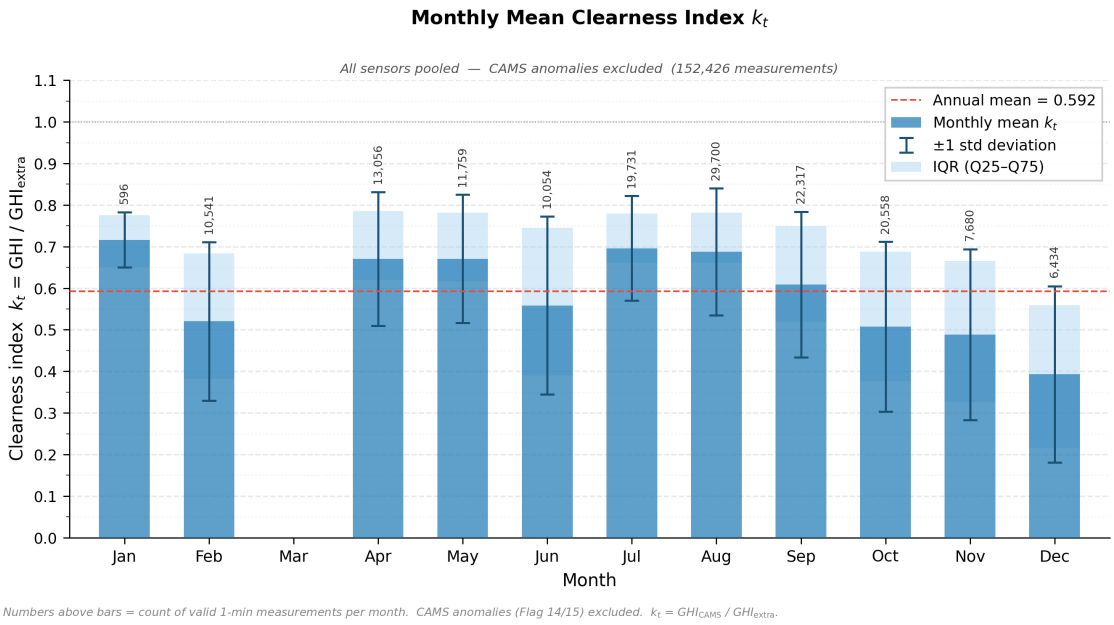


**Figure B.6:** Monthly mean clearness index, standard campaign, hq variant. It peaks in July and August (0.695 and 0.685) and collapses in December (0.380); March is absent and the January value rests on 630 minutes.

# Appendix C

# Data preparation method

## C.1 Principle

Every stage of the preparation takes one CSV file and writes another. No stage modifies its input, and each appends its own segment to the name it received. A file therefore carries its own history, and two files whose names differ by one segment differ by exactly one treatment.

Two consequences follow, and they are the reason for the convention. Any analysis script can be pointed at a different stage of the chain by changing a single parameter, since the path is reconstructed from the campaign, the sensor and the active suffix. And a figure can always be traced back to the exact subset that produced it, without an external record, because the subset is named in the file it came from.

## C.2 The two join chains

The joins run before any cleaning, so that each spatial computation is performed once rather than once per quality level. The two campaigns keep separate chains because their raw exports differ, and they converge on a common column schema before the cleaning stage (Table C.1). Every join appends columns only: with a single exception, noted below, none of them changes the number of rows.

For performance, the spatial computation in each join is restricted to rows already known to be valid: non-null coordinates, a non-zero calibrated reading, and a position inside mainland Portugal. That last test also catches the lat $=$ lon $=$ 0 sentinel, which a null check would let through. The remaining rows receive an explicit missing marker rather than a silent blank.

The cleaning stage is the only mandatory one, and it produces a family of variants rather than a single file; Figure 2.1 gives that family, and Section 2.3 the criterion applied at each step. The two geographic filters are alternatives, not successive stages.

## C.3 Construction of the tuk-tuk files

This chain deserves a description of its own, because it is the one that the continuation of the campaign will use, and because it starts much further upstream than the standard one. The J30–J33 export used in Chapter 3 arrived with irradiance, solar position and land use already attached; the tuk-tuk export is raw device telemetry, so each of those has to be constructed explicitly. Table C.2 lists what every stage contributes.

**Table C.1:** Order of the preparation stages and the file each produces, abbreviated to the last two segments of the name. The standard chain starts from an export in which irradiance, solar position and land use are already present; the tuk-tuk chain has to build them, which is why it is longer. Figure C.1 gives the names in full.

| Stage | Resulting file |
|---|---|
| *Standard campaign* | |
| — | `j3x_satellite_landuse_v2.csv` |
| `join_cos_standard.py` | `…_v2_cos.csv` |
| `join_bh_standard.py` | `…_cos_bh.csv` |
| `join_sw_standard.py` | `…_bh_sw.csv` |
| `join_clc_standard.py` | `…_sw_clc.csv` |
| *Tuk-tuk campaign* | |
| `decode_tuktuk.py` | `j3x_tuktuk.csv` |
| `join_ghi_fcul.py` | `…_tuktuk_ghi.csv` |
| `join_cams_tuktuk.py` | `…_ghi_cams.csv` |
| `join_cos23_tuktuk.py` | `…_cams_cos.csv` |
| `join_clc_tuktuk.py` | `…_cos_clc.csv` |
| `join_bh_tuktuk.py` | `…_clc_bh.csv` |
| `join_sw_tuktuk.py` | `…_bh_sw.csv` |
| `normalize_tuktuk.py` | `…_sw_norm.csv` |
| *Both campaigns* | |
| `clean_data.py` | `…_sd_cleaned[_hq][_clearsky].csv` |
| `filter_lisbon.py` *or* `filter_region.py` | `…_lisbon.csv` / `…_region.csv` |
| `select_time_range.py` | `…_<label>.csv` |

**Decoding.** The database export stores one JSON payload per transmitted packet, each holding up to five measurements in delta-encoded arrays. The first element of an array is the absolute value; the others are offsets from that first element, not from their predecessor. Decoding therefore reconstructs $v_i = v_0 + \delta_i$ rather than accumulating differences, a distinction that silently corrupts the series if it is read the other way. Rows without GPS coordinates are dropped, since a measurement with no position has no use in a spatial analysis, as are rows whose timestamp falls outside the campaign window, which removes packets carrying a corrupt clock value. The calibrated column is written at this stage as rad_calib $= a \cdot A_0 + b$; the coefficients are per sensor and currently $a = 1$, $b = 0$, so the two radiometric columns are identical until the calibration session takes place.

**The ground reference, and the one stage that removes rows.** The FCUL rooftop pyranometer is joined first, by linear interpolation in time onto each measurement timestamp, without extrapolation beyond the range covered by the reference. This campaign is confined to Lisbon, close to that instrument, which is what makes a ground reference preferable to a satellite one here. This stage is also the only one in the chain that changes the number of rows: packets re-sent to the database appear twice with the same measurement timestamp and different storage times, and the duplicates are removed here. The row count is therefore fixed at this point, and every file downstream can be compared to it directly.

**Irradiance components and solar position.** The CAMS join then attaches the six radiation components and the two solar angles. Each measurement is snapped to the nearest node of a 0.05° grid for the download, but the solar position is computed with `pvlib` at the true GPS coordinates, not at the node: the grid is a property of the radiation product, not of the geometry. Requests are pooled across sensors and cached locally, because the number of daily requests to the service is capped and the same node serves many measurements.

**Spatial context.** The four remaining joins are the same computations as their standard-campaign counterparts, differing only in the column names they read: the tuk-tuk files still carry generic `lat` and `lon` at this point, where the standard files carry `J31 lat` and `J31 lon`.

**Normalisation, last.** Renaming to the pipeline column format is the final step of the chain, not the first. Running it last means every join script reads the decoder's own column names, and only one file per sensor has to be translated. That file is the convergence point: from there on, a tuk-tuk file and a standard file have the same column schema and are handled by exactly the same code.

**Table C.2:** Columns contributed by each stage of the tuk-tuk chain.

| Stage | Columns added |
|---|---|
| `decode_tuktuk` | `timestamp`, `lat`, `lon`, `A0`, `rad_calib`, `voltage_V`, `temp_C` |
| `join_ghi_fcul` | `ghi_fcul` |
| `join_cams` | `ghi`, `ghi_clear`, `ghi_extra`, `bhi`, `dhi`, `dni`, `sun_elev`, `sun_azim` |
| `join_cos23` | `cos23_n4_code`, `cos23_n4_label`, `cos23_n1_label` |
| `join_clc` | `land_use_clcv`, `land_use_str_clcv` (`clc_urban_rural` on the tuk-tuk chain only) |
| `join_bh` | `mean_building_height_m`, `max_building_height_m`, `n_buildings_nearby` |
| `join_sw` | `street_width_m`, `width_source`, `edge_match_dist_m` |
| `normalize` | none, renames only |

## C.4 How a name is assembled

Figure C.1 follows both chains segment by segment. Each row adds one segment, highlighted, and names the script responsible for it.

The two panels join at the cleaning stage, but they do not enter it the same way: the cleaning script builds its output name from a stem fixed per campaign rather than from the name of the file it reads. For the standard campaign that stem is the raw export, so the four join segments are dropped: the joins are performed once and their columns are present everywhere downstream, so repeating `_cos_bh_sw_clc` in every name would add length without adding information. For the tuk-tuk campaign the stem is the full normalised name, kept whole. The names are consequently much longer, and the asymmetry is defensible rather than accidental: the standard join chain is fixed and always

runs in full, so its segments carry no information, whereas the tuk-tuk chain constructs the irradiance, solar-position and land-cover columns itself, and keeping its segments records which sources were actually attached to a given file.

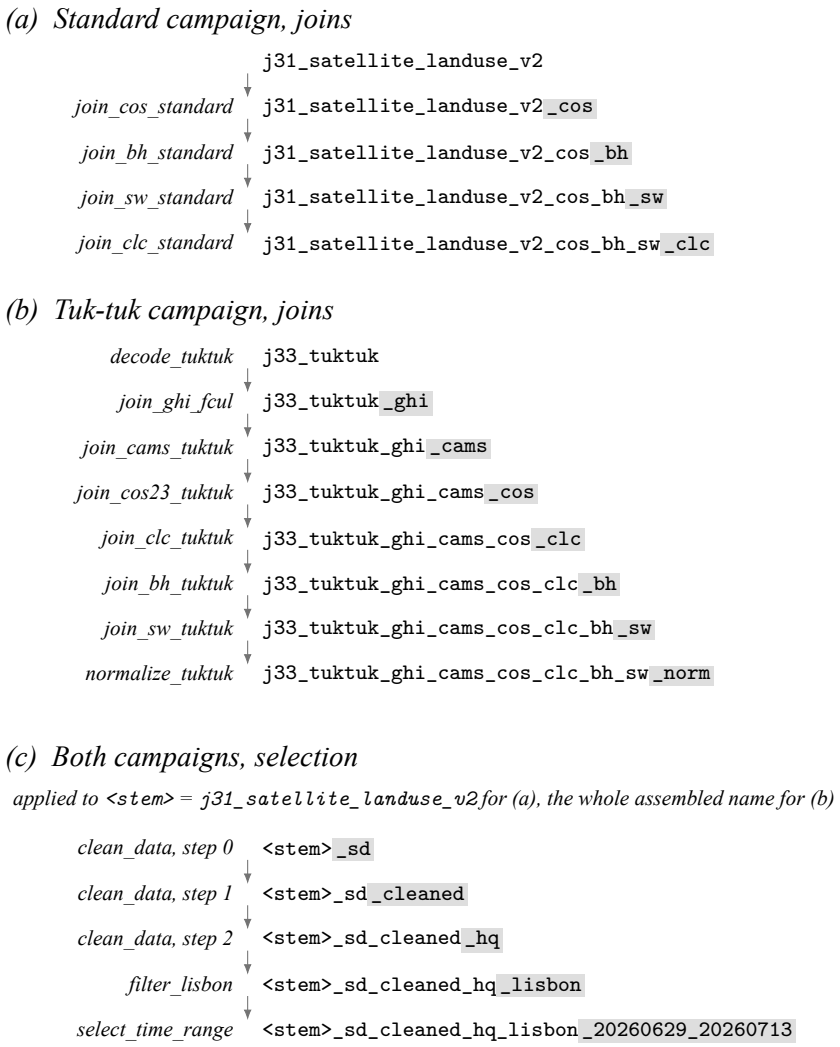


**Figure C.1:** Assembly of a file name along each preparation chain. The segment added at each stage is highlighted, and the script responsible is named on the left. The selection stages in (c) are shared, but the stem they build on is not: the standard chain drops its join segments, the tuk-tuk chain keeps them.

## C.5 Reading a name

Table C.3 gives the meaning of each segment. Read in order, they answer four questions: which sensor, which quality level, which geographic extent, and which period. A name that omits a segment omits the corresponding restriction: an absent `_hq` means the quality filter was not applied, not that it failed.

**Table C.3:** Filename suffix segments and the pipeline stage each records. Segments are appended left to right in the order the scripts are run.

| Segment | Example | Stage |
|---|---|---|
| `jxx` | `j31` | Sensor identifier |
| `_sd` | `_sd` | Start-date filter, excluding the rooftop calibration phase |
| `_cleaned` | `_cleaned` | Valid rows, the mandatory base of every variant |
| `_hq` | `_hq` | High-quality radiometric and geometric filter |
| `_clearsky` | `_clearsky` | Clear-sky filter, $\mathrm{GHI}/\mathrm{GHI}_{\mathrm{clear}} \geq 0.95$ |
| `_lisbon` | `_lisbon` | Strict Lisbon municipality boundary |
| `_region` | `_region` | Lisbon–Cascais bounding box |
| `_<label>` | `_20220901_20220930` | Time-range subset |

# Appendix D

# CORINE Land Cover classification

CORINE Land Cover is the pan-European land-cover product of the Copernicus Land Monitoring Service, mapped at 100 m with a minimum mapping unit of 25 ha and a minimum width of 100 m for linear features. Its legend is hierarchical: five level-1 groups divide into fifteen level-2 groups and then into the 44 level-3 classes reproduced in Table D.1. This work uses it as the secondary classification, kept because it makes the results comparable outside Portugal (Chapter E covers the primary one).

One point matters for anyone reusing the data files, because two generations of land-cover assignment coexist in them. The raw sensor exports already carried a `land_use` column and a `land_use_str` label, produced before this pipeline existed by a raster lookup whose provenance is not documented. Neither is used here. They are partially wrong, and the string column is offset by one class throughout, so it does not even label its own numeric column correctly. The columns are left in the files rather than deleted, but nothing reads them.

The reference used throughout this work is `land_use_clcv`, computed by `join_clc` as a point-in-polygon test against the CLC 2018 vector product, taking the `Code_18` attribute of the polygon containing the point and mapping it to the grid code of Table D.1. A second recomputation exists, `land_use_clcr`, which samples the 100 m CLC 2018 raster at the same coordinates. It is not used in any analysis: its purpose was to check the vector one against an independent method. The two agree on 98 % to 99 % of points, and on a wall-to-wall comparison over western Lisbon they are pixel-identical, 0 disagreements over 6,268 pixels. The legacy column agrees with neither as consistently, which is what disqualified it.

Where it fails is not random. Against the vector reference it agrees on 96.9 % of J30 points but only 81.9 % of J31 points, and 74 % of all disagreements are a single substitution: class 21, land principally occupied by agriculture with significant areas of natural vegetation, where the vector product reads class 2, discontinuous urban fabric. That pair is the signature of the 25 ha minimum mapping unit: a vegetated patch embedded in urban fabric below that size is absorbed into the surrounding urban polygon by design, so the two sources part company exactly at the edges of built-up areas, which is where a shading study does most of its work.

Values of −999, marking a point outside every polygon, and anything outside 1 to 44 are removed during cleaning, which filters on `land_use_clcv`. The urban and rural split is not stored as a column at all: each script derives it in memory from `land_use_clcv` through the last column of Table D.1.

That aggregation splits the 44 classes into 9 urban and 35 rural. It follows level 1 with two exceptions: the urban side is the artificial-surfaces group minus mineral extraction sites and dump sites, which are artificial but open, and everything in the four remaining groups is rural. Construction sites,

green urban areas and sport and leisure facilities therefore count as urban, although two of the three are open ground. This is a choice rather than a property of the nomenclature, and Section 3.5 reports what the comparison it supports was, and was not, able to establish.

**Table D.1:** The 44 CORINE Land Cover level-3 classes, grouped by level-1 and level-2 heading. The first column is the value stored in the `land_use` column of the data files.

| Grid | CLC | Level-3 class | Aggregation |
|---|---|---|---|
| **1** | **Artificial surfaces** | | |
| *11* | *Urban fabric* | | |
| 1 | 111 | Continuous urban fabric | Urban |
| 2 | 112 | Discontinuous urban fabric | Urban |
| *12* | *Industrial, commercial and transport units* | | |
| 3 | 121 | Industrial or commercial units | Urban |
| 4 | 122 | Road and rail networks and associated land | Urban |
| 5 | 123 | Port areas | Urban |
| 6 | 124 | Airports | Urban |
| *13* | *Mine, dump and construction sites* | | |
| 7 | 131 | Mineral extraction sites | Rural |
| 8 | 132 | Dump sites | Rural |
| 9 | 133 | Construction sites | Urban |
| *14* | *Artificial, non-agricultural vegetated areas* | | |
| 10 | 141 | Green urban areas | Urban |
| 11 | 142 | Sport and leisure facilities | Urban |
| **2** | **Agricultural areas** | | |
| *21* | *Arable land* | | |
| 12 | 211 | Non-irrigated arable land | Rural |
| 13 | 212 | Permanently irrigated land | Rural |
| 14 | 213 | Rice fields | Rural |
| *22* | *Permanent crops* | | |
| 15 | 221 | Vineyards | Rural |
| 16 | 222 | Fruit trees and berry plantations | Rural |
| 17 | 223 | Olive groves | Rural |
| *23* | *Pastures* | | |
| 18 | 231 | Pastures | Rural |
| *24* | *Heterogeneous agricultural areas* | | |
| 19 | 241 | Annual crops associated with permanent crops | Rural |
| 20 | 242 | Complex cultivation patterns | Rural |
| 21 | 243 | Land principally occupied by agriculture, with significant areas of natural vegetation | Rural |

*continued on the next page*

*Table D.1, continued*

| Grid | CLC | Level-3 class | Aggregation |
|---|---|---|---|
| 22 | 244 | Agro-forestry areas | Rural |
| **3** | **Forest and semi-natural areas** | | |
| *31* | *Forests* | | |
| 23 | 311 | Broad-leaved forest | Rural |
| 24 | 312 | Coniferous forest | Rural |
| 25 | 313 | Mixed forest | Rural |
| *32* | *Shrub and/or herbaceous vegetation associations* | | |
| 26 | 321 | Natural grasslands | Rural |
| 27 | 322 | Moors and heathland | Rural |
| 28 | 323 | Sclerophyllous vegetation | Rural |
| 29 | 324 | Transitional woodland-shrub | Rural |
| *33* | *Open spaces with little or no vegetation* | | |
| 30 | 331 | Beaches, dunes, sands | Rural |
| 31 | 332 | Bare rocks | Rural |
| 32 | 333 | Sparsely vegetated areas | Rural |
| 33 | 334 | Burnt areas | Rural |
| 34 | 335 | Glaciers and perpetual snow | Rural |
| **4** | **Wetlands** | | |
| *41* | *Inland wetlands* | | |
| 35 | 411 | Inland marshes | Rural |
| 36 | 412 | Peat bogs | Rural |
| *42* | *Maritime wetlands* | | |
| 37 | 421 | Salt marshes | Rural |
| 38 | 422 | Salines | Rural |
| 39 | 423 | Intertidal flats | Rural |
| **5** | **Water bodies** | | |
| *51* | *Inland waters* | | |
| 40 | 511 | Water courses | Rural |
| 41 | 512 | Water bodies | Rural |
| *52* | *Marine waters* | | |
| 42 | 521 | Coastal lagoons | Rural |
| 43 | 522 | Estuaries | Rural |
| 44 | 523 | Sea and ocean | Rural |

# Appendix E

# COS2023 classification

COS2023 (*Carta de Uso e Ocupação do Solo*, série 2) is the national land-cover map of Portugal, published by the Direção-Geral do Território. It is a vector product with a minimum mapping unit of 1 ha and a minimum polygon width of 20 m, against 25 ha and 100 m for CORINE (Chapter D), and that difference is why it is the primary classification here: at 25 ha a city block dissolves into its surroundings, at 1 ha it does not.

The legend has nine level-1 branches and 93 level-4 classes, listed in Table E.1 with the official English labels published by the DGT. The distribution is uneven and reflects the purpose of the product: 37 of the 93 classes describe artificial surfaces, against a single class for shrubland.

The class that matters most to this work is the pair `1.1.1.1` and `1.1.1.2`, continuous residential fabric predominantly vertical and predominantly horizontal. COS2023 is the only one of the two nomenclatures that encodes the verticality of the built fabric, which is the variable a shading study needs and which CORINE collapses into a single continuous urban class.

Unlike CORINE, COS2023 carries no urban and rural aggregation in this work. The two-way split used in the results chapter is therefore always the CORINE one, and the COS2023 analysis stays at class level.

**Table E.1:** The 93 COS2023 level-4 classes, grouped by level-1 branch. Labels are the official English translations published by the DGT.

| Code | Class |
|---|---|
| **1** | **Artificial surfaces** |
| 1.1.1.1 | Continuous residential built-up areas, predominantly vertical |
| 1.1.1.2 | Continuous residential built-up areas, predominantly horizontal |
| 1.1.2.1 | Discontinuous residential built-up areas |
| 1.1.2.2 | Sparse discontinuous residential built-up areas |
| 1.2.1.1 | Industry and logistics |
| 1.2.1.2 | Commerce and services |
| 1.2.2.1 | Agricultural and livestock facilities |
| 1.3.1.1 | Cultural facilities |
| 1.3.2.1 | Sport facilities |
| 1.3.2.2 | Leisure facilities |
| 1.3.2.3 | Golf courses |

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*Table E.1, continued*

| Code | Class |
|---|---|
| 1.3.2.4 | Camping and caravan sites |
| 1.3.3.1 | Cemeteries |
| 1.3.4.1 | Other tourist facilities and equipment |
| 1.4.1.1 | Hydropower production infrastructures |
| 1.4.1.2 | Solar energy production infrastructures |
| 1.4.1.3 | Other renewable energy production infrastructures |
| 1.4.2.1 | Fossil fuel energy production infrastructures |
| 1.4.3.1 | Substations and energy transformation stations |
| 1.4.4.1 | Water extraction and treatment infrastructures for consumption |
| 1.4.4.2 | Wastewater drainage and treatment infrastructures |
| 1.4.5.1 | Landfills |
| 1.4.5.2 | Other waste infrastructures |
| 1.4.6.1 | Other infrastructures |
| 1.5.1.1 | Road network |
| 1.5.1.2 | Rail network |
| 1.5.2.1 | Sea and river port terminals |
| 1.5.2.2 | Shipyards and dry docks |
| 1.5.2.3 | Marinas and fishing docks |
| 1.5.3.1 | Airports |
| 1.5.3.2 | Airfields |
| 1.5.4.1 | Parking areas |
| 1.6.1.1 | Open-cast mines |
| 1.6.1.2 | Quarries |
| 1.7.1.1 | Unbuilt vacant spaces |
| 1.7.1.2 | Construction sites |
| 1.8.1.1 | Green spaces |
| **2** | **Agricultural areas** |
| 2.1.1.1 | Rainfed and irrigated temporary crops |
| 2.1.1.2 | Rice fields |
| 2.2.1.1 | Vineyards |
| 2.2.2.1 | Orchards |
| 2.2.3.1 | Olive groves |
| 2.3.1.1 | Temporary crops and/or improved pastures associated with vineyards |
| 2.3.1.2 | Temporary crops and/or improved pastures associated with orchards |
| 2.3.1.3 | Temporary crops and/or improved pastures associated with olive groves |
| 2.3.2.1 | Complex cultivation and parcel mosaics |
| 2.3.3.1 | Agriculture with natural and semi-natural spaces |

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*Table E.1, continued*

| Code | Class |
|---|---|
| 2.4.1.1 | Protected agriculture and greenhouses |
| **3** | **Pastures and natural grasslands** |
| 3.1.1.1 | Improved pastures |
| 3.1.2.1 | Natural grasslands / Spontaneous pastures |
| **4** | **Agro-forestry surfaces** |
| 4.1.1.1 | Agro-silvicultural surfaces with cork oak |
| 4.1.1.2 | Agro-silvicultural surfaces with holm oak |
| 4.1.1.3 | Agro-silvicultural surfaces with other oaks |
| 4.1.1.4 | Agro-silvicultural surfaces with other broad-leaved trees |
| 4.1.2.1 | Agro-silvicultural surfaces with stone pine |
| 4.1.2.2 | Agro-silvicultural surfaces with other conifers |
| 4.2.1.1 | Silvopastoral surfaces with cork oak |
| 4.2.1.2 | Silvopastoral surfaces with holm oak |
| 4.2.1.3 | Silvopastoral surfaces with other oaks |
| 4.2.1.4 | Silvopastoral surfaces with other broad-leaved trees |
| 4.2.2.1 | Silvopastoral surfaces with stone pine |
| 4.2.2.2 | Silvopastoral surfaces with other conifers |
| **5** | **Forests** |
| 5.1.1.1 | Cork oak forests |
| 5.1.1.2 | Holm oak forests |
| 5.1.1.3 | Other oak forests |
| 5.1.1.4 | Chestnut forests |
| 5.1.1.5 | Carob forests |
| 5.1.1.6 | Eucalyptus forests |
| 5.1.1.7 | Acacia forests |
| 5.1.1.8 | Other broad-leaved forests |
| 5.1.2.1 | Maritime pine forests |
| 5.1.2.2 | Stone pine forests |
| 5.1.2.3 | Other coniferous forests |
| **6** | **Shrublands** |
| 6.1.1.1 | Shrublands |
| **7** | **Open spaces with little or no vegetation** |
| 7.1.1.1 | Inland beaches, dunes and sands |
| 7.1.1.2 | Coastal beaches, dunes and sands |
| 7.1.2.1 | Rocky areas |

*continued on the next page*

*Table E.1, continued*

| Code | Class |
|---|---|
| 7.1.3.1 | Sparsely vegetated areas |
| **8** | **Wetlands** |
| 8.1.1.1 | Peat bogs and fens |
| 8.1.2.1 | Salt marshes |
| 8.1.2.2 | Intertidal zones |
| **9** | **Surface water bodies** |
| 9.1.1.1 | Natural watercourses |
| 9.1.1.2 | Modified or artificial watercourses |
| 9.1.2.1 | Artificial inland lakes and ponds |
| 9.1.2.2 | Natural inland lakes and ponds |
| 9.1.2.3 | Dam reservoirs |
| 9.1.2.4 | Weir reservoirs |
| 9.1.2.5 | Ponds |
| 9.2.1.1 | Aquaculture |
| 9.3.1.1 | Salinas |
| 9.3.2.1 | Coastal lagoons |
| 9.3.3.1 | Estuaries / River mouths |
| 9.3.4.1 | Ocean |

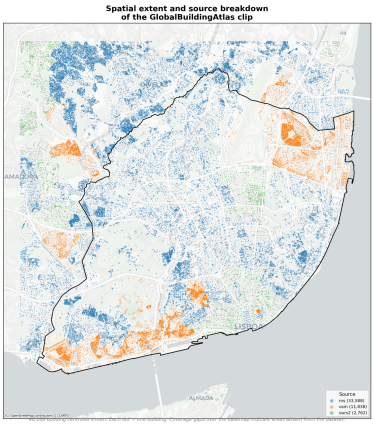
# Appendix F

# Building-height coverage

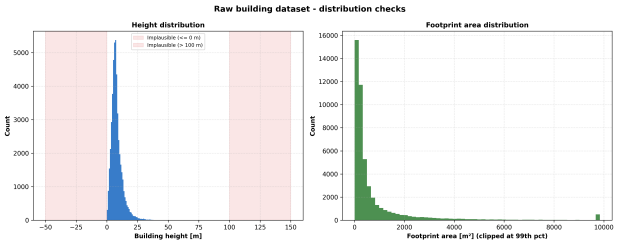
Building height is the predictor that survives every filter in Section 3.7, so what the join does and does not cover decides how far that result reaches. Figure F.1 gathers the six panels of the diagnostic written for it, on the Lisbon extent.

The number to keep is the zero-building rate. Of the 84,750 measurement points on this extent, 21.0 % find no building centroid within the 50 m buffer, and that absence is not spread evenly: panel (e) puts it above 80 % for grassland, shrubland, airports and stone-pine forest, and below 10 % for the dense residential classes. A zero is a legitimate reading, an open space or a wide road, and is kept distinct from a missing value, but it means the height axis is measured where buildings are and is silent elsewhere.

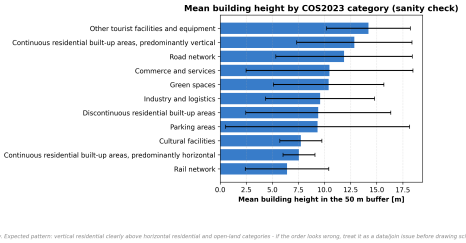
Panel (c) is the sanity check the method rests on. Continuous residential fabric of predominantly vertical morphology comes out around 12.9 m against 7.5 m for the predominantly horizontal one, which is the ordering the COS2023 nomenclature asserts independently. Had the two come out level, the joined heights would have been a data problem rather than a result.



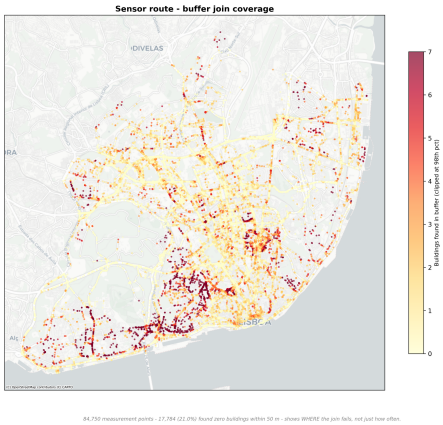

**(a)** The 48,188 GlobalBuildingAtlas centroids of the clip, one dot per building, coloured by the source the footprint comes from. Blanks over the basemap are buildings the dataset does not hold.



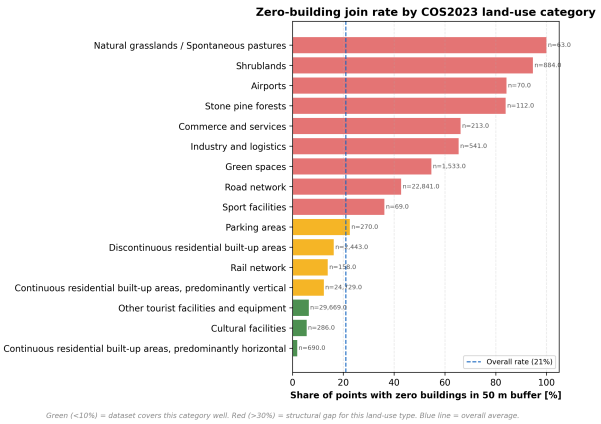

**(b)** Distribution of the raw heights and footpring areas, with the implausible bands, at or below 0 m and above 100 m, shaded. Neither band is populated here.



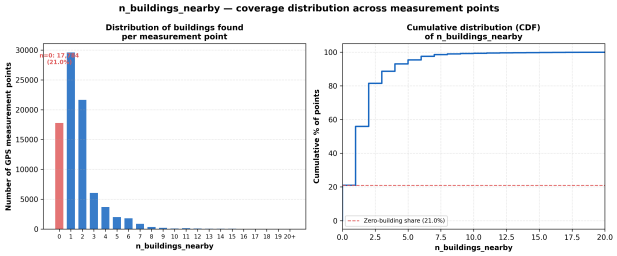

**(c)** Mean joined height by COS2023 class, with its standard deviation. Vertical residential fabric sits clearly above the horizontal one, as the nomenclature asserts.




**(d)** The driven route coloured by the number of buildings found in the buffer: where the join is dense, where it is thin, and where it returns nothing.




**(e)** Share of points finding zero buildings in the buffer, by COS2023 class. The vertical line is the overall 21 %.




**(f)** How many buildings each point finds, and the cumulative distribution. Four points in five find between zero and two, so the mean height often rests on a very small sample.

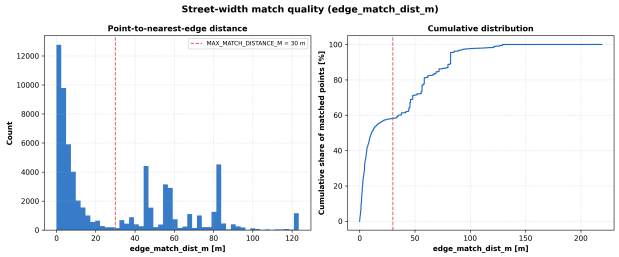
**Figure F.1:** Coverage and quality of the GlobalBuildingAtlas building-height join, standard campaign, `sd_cleaned_lisbon` variant. The panels are reproduced at the size of the grid; each is readable at full size in the output folder of `building_height_quality_diagnostic.py`.

# Appendix G

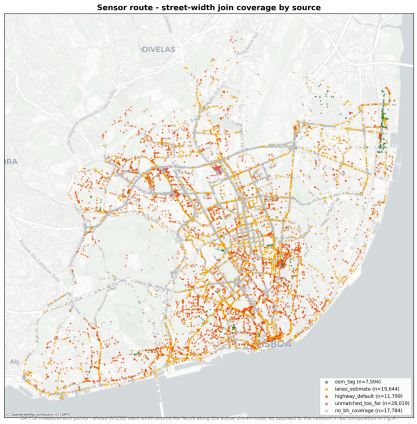
# Street-width coverage

Street width is the one input of this work that is not a downloaded file. It is read from the OpenStreetMap network through the Overpass API at run time, and its quality varies from street to street rather than being a property of the dataset as a whole. Figure G.1 gathers the six panels of the diagnostic written for it, on the Lisbon extent, the only one over which the network was retrieved. The extent is itself restricted to the points that already carry a valid building height, since a width is only ever used once $H$ is known, to form the ratio $H/W$ of Section 2.5.4.

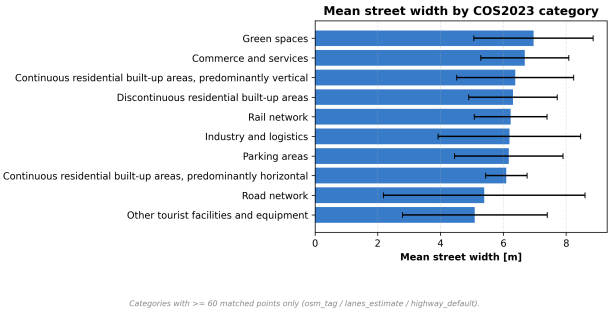
Two things are worth reading off the grid. The provenance of a width is not uniform: an explicit `width` tag, a lane count and a road-class default do not carry the same authority, and panels (b) and (d) show where each of the three lands, on the network and along the driven route. And the join fails unevenly across land-cover classes, panel (f), which is what makes the failure a structural gap for some kinds of street rather than a uniform thinning of the sample.



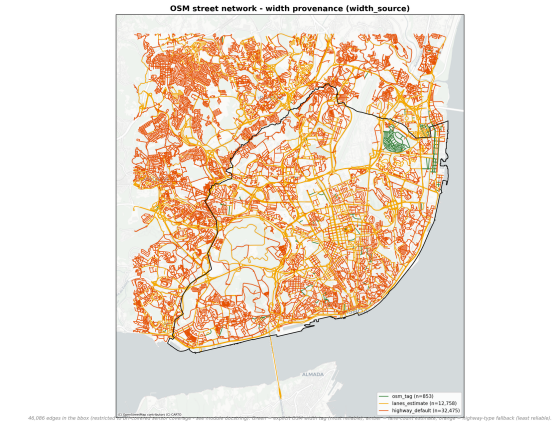

**(a)** Distance from each point to the nearest street edge, and its cumulative distribution. The dashed line is the maximum match distance beyond which a point is left unmatched.



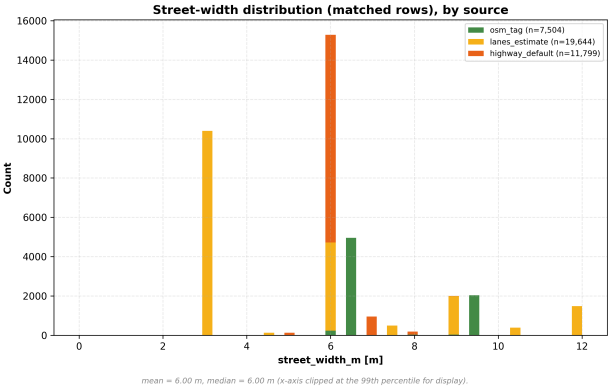

**(b)** The driven route coloured by the source of the width attached to each point: where each of the three tiers actually lands.



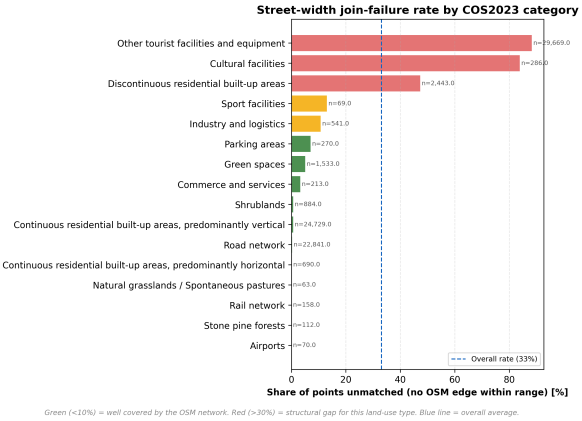

**(c)** Mean street width by COS2023 class, with its standard deviation. Classes holding fewer than 60 matched points are omitted.




**(d)** The OSM network itself over the same extent, coloured by width provenance: explicit tag, lane-count estimate, or road-class fallback.




**(e)** Distribution of the matched widths, stacked by source. The axis is clipped at the 95th percentile for display.




**(f)** Share of points for which no street edge was found within range, by COS2023 class. The vertical line is the overall rate.

**Figure G.1:** Coverage and quality of the OpenStreetMap street-width join, standard campaign, `sd_cleaned_lisbon` variant. The panels are reproduced at the size of the grid; each is readable at full size in the output folder of `street_width_quality_diagnostic.py`.

# Appendix H

# Tuk-tuk campaign poster

The campaign puts a research instrument on a vehicle carrying paying passengers, which makes the sensor visible to people. The notice below, displayed on board throughout, states what the box on the roof records, who is running the study and why, and carries a QR code to the project page.

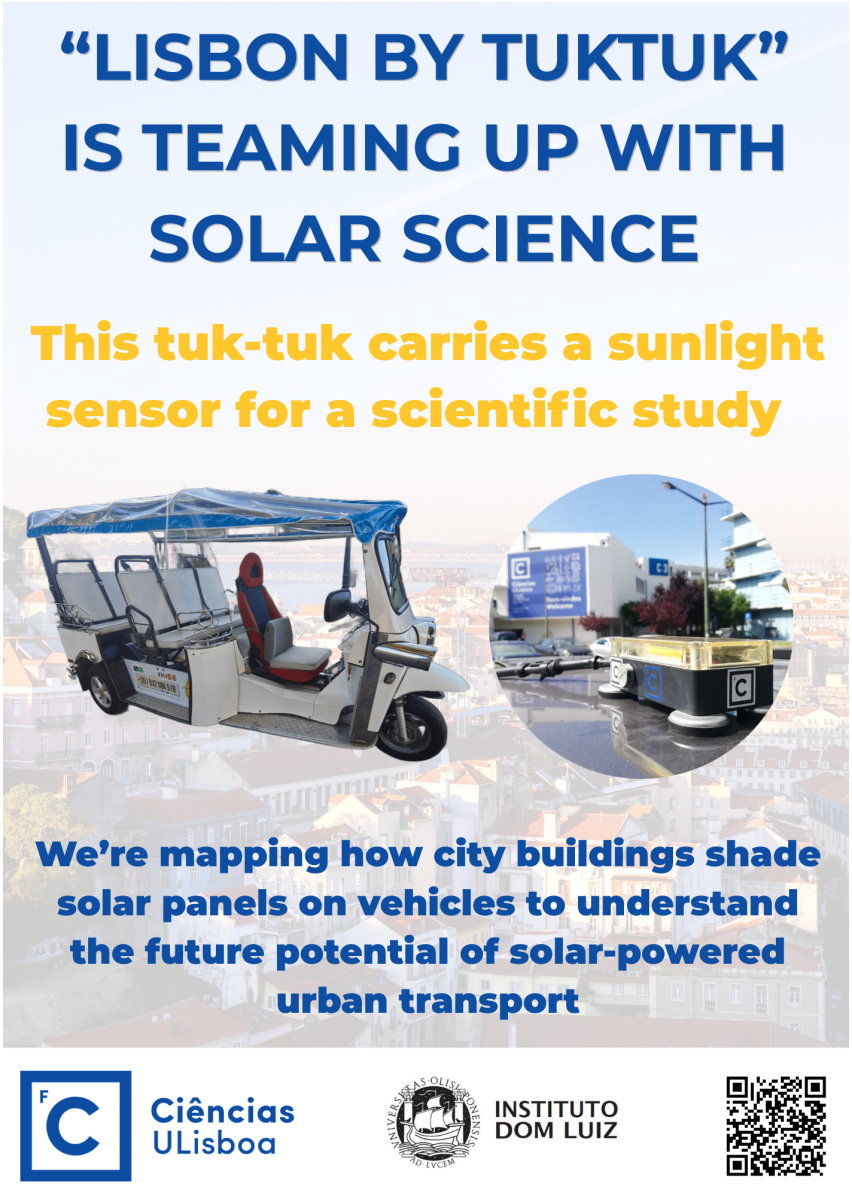



**Figure H.1:** Notice displayed in the instrumented tuk-tuk. The inset photograph shows the Solar Jinie sensor as magnetically fixed to the roof.